Reconocimiento de Patrones

Versión 2026

Características Geométricas

(CAPÍTULO II – PR02A)

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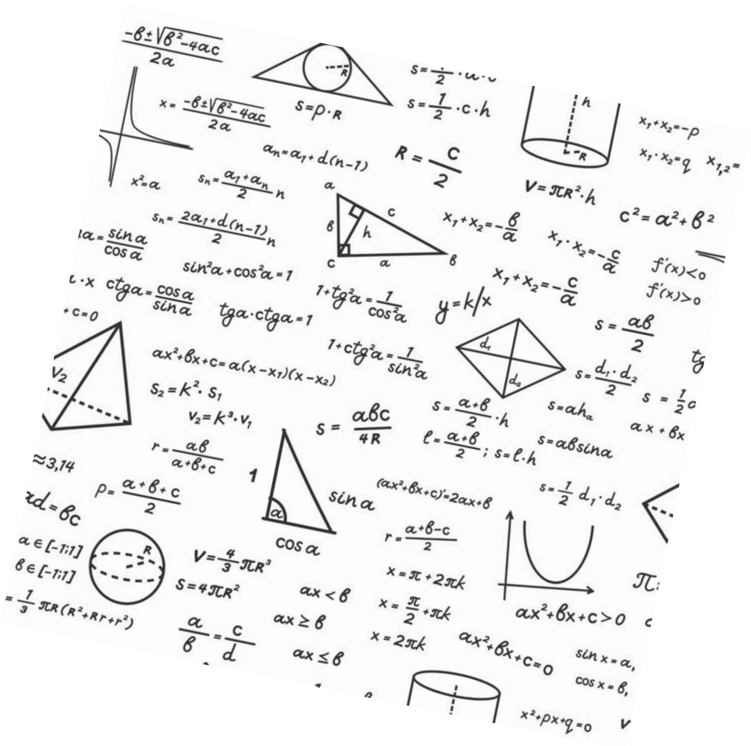
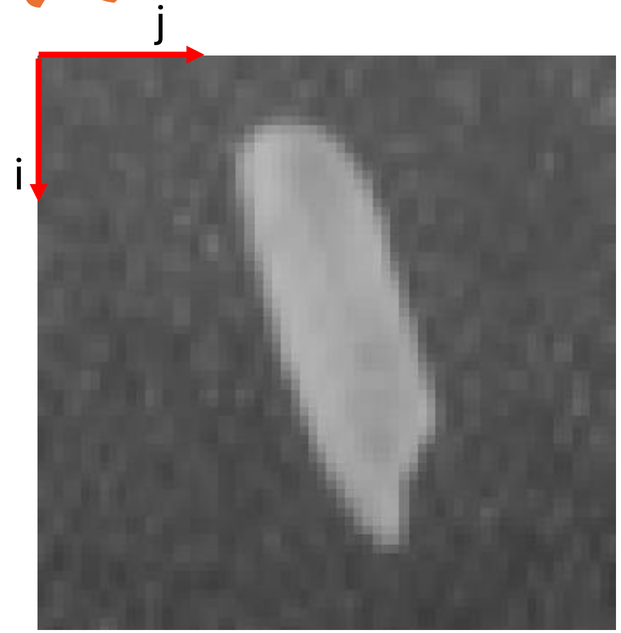
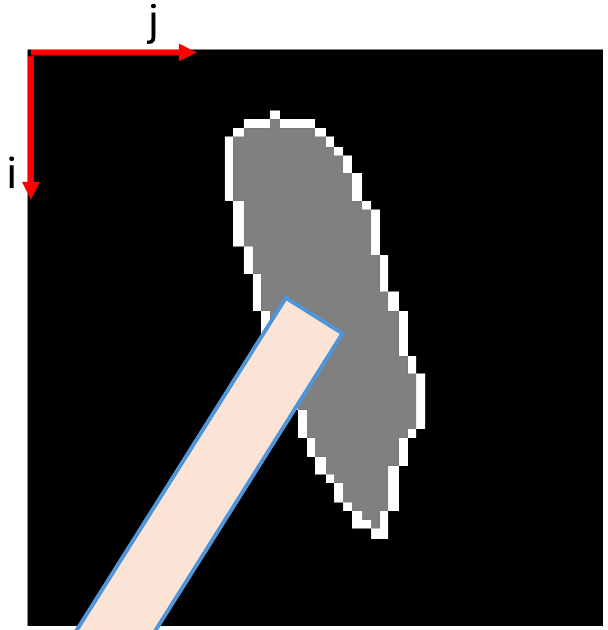


Introducción

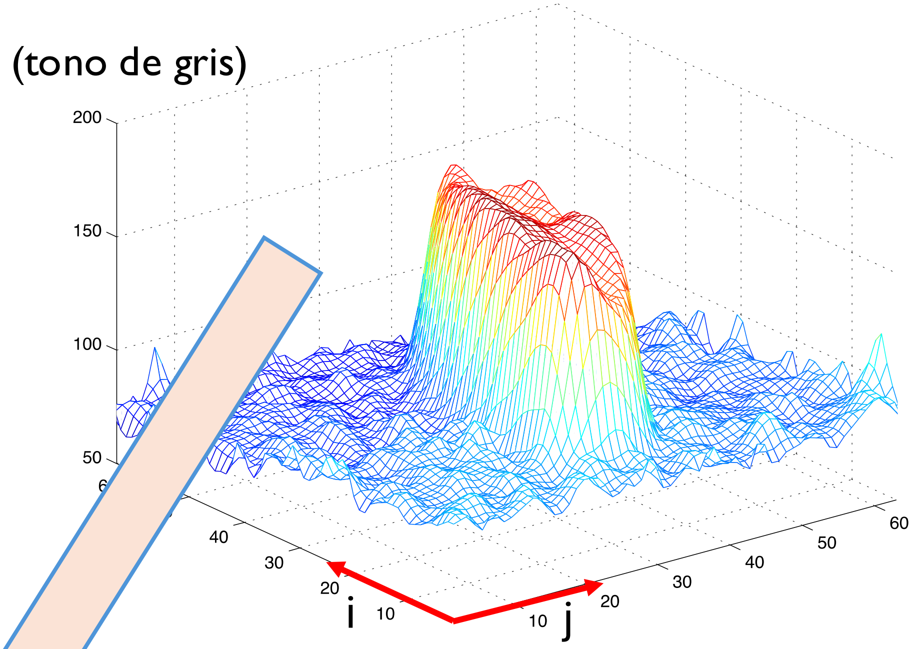
¿Qué es la Extracción de Características?



a) Imagen en escala de grises



b) Segmentación



c) Representación 3D de a)

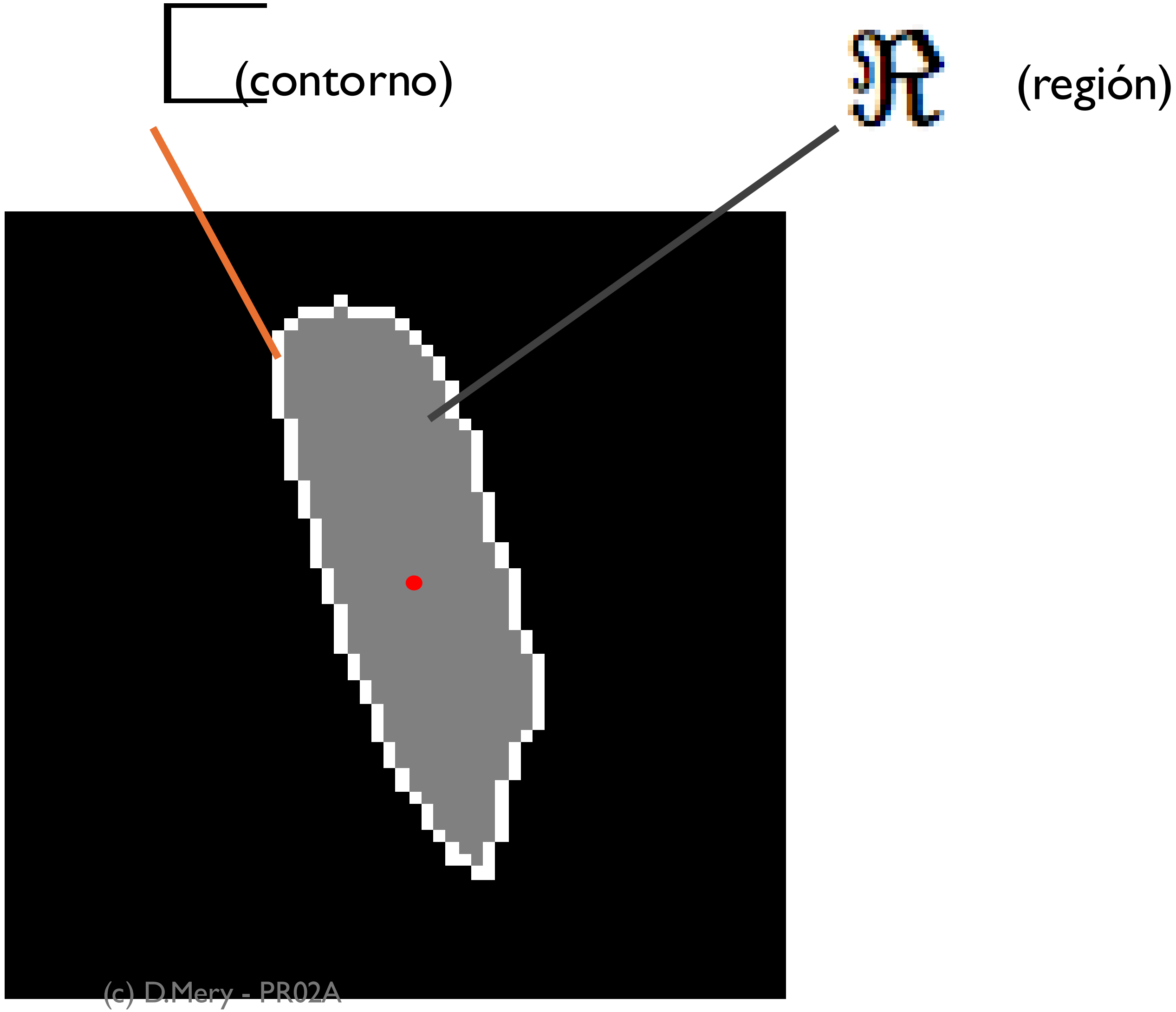
Existen dos categorías de características:

1) Las Características Geométricas proporcionan información sobre traslación, rotación, forma y tamaño.

2) Las Características de Intensidad proporcionan información sobre cómo se distribuyen los tonos de grises (o los canales de color).

Características geométricas

Características Básicas



Área y perímetro

A = # de píxeles grises

L = # de píxeles blancos

Altura y ancho de $\Re$

$$h = i_{max} - i_{min} + 1$$

$$w = j_{max} - j_{min} + 1$$

Redondez

$$R = \frac{4 \cdot A \cdot \pi}{L^2}$$

Centro de masa

(i_m, j_m)

Ejemplo: imagen binaria 9x9 – Características Básicas

	1	2	3	4	5	6	7	8	9
1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	1	1	0	0
5	0	0	0	0	1	1	0	0	0
6	0	0	0	1	1	0	0	0	0
7	0	0	1	1	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0

j

i

En la región R:

$$i = \{4,4,5,5,6,6,7,7\}$$

$$j = \{3,4,4,5,5,6,6,7\}$$

$$h = i_{\max} - i_{\min} + 1 = 4$$

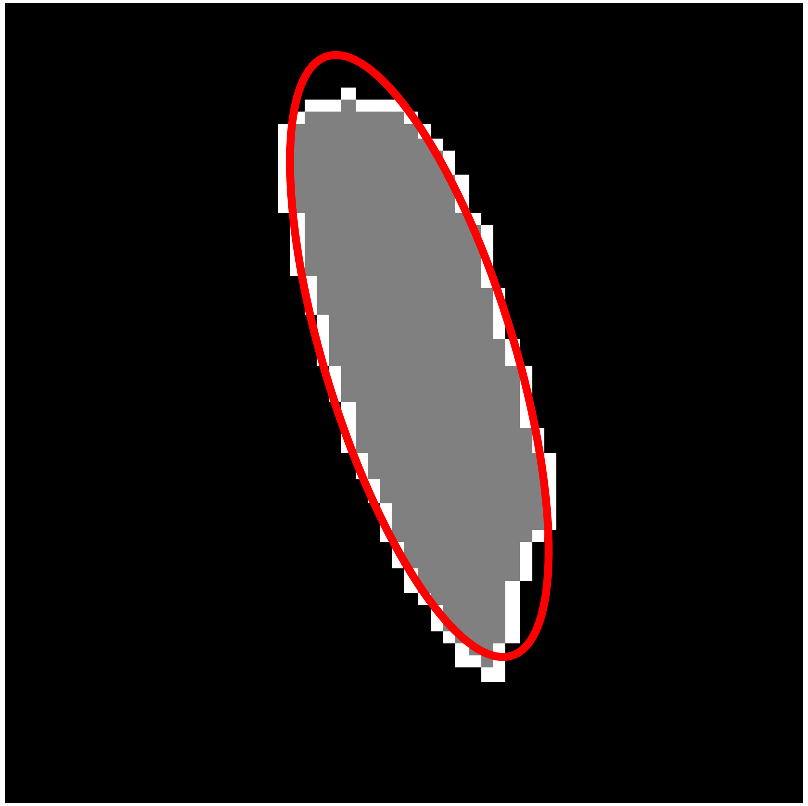
$$w = j_{\max} - j_{\min} + 1 = 5$$

$$A = 8$$

$$i_m = \frac{1}{A}\sum i = 5.5$$

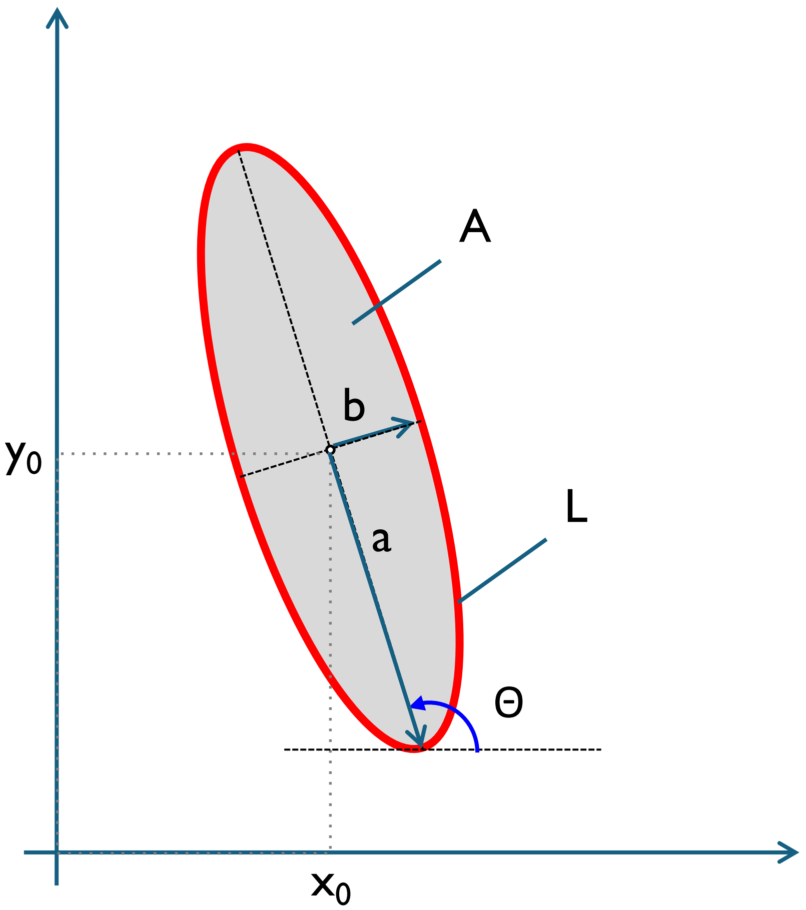
$$j_m = \frac{1}{A}\sum j = 5.0$$

Elipses



Elipses

Eje mayor	(a)
Eje menor	(b)
Orientación	(θ)
Centro	x_0, y_0
Área	A
Perímetro	L
Excentricidad	E = b/a



Momentos

$$m_{rs} = \sum_{i,j \in \Re} i^r j^s \qquad \text{for } r,s \in \mathbb{N}$$

m_{00}

(área)

(centro de masa: i_m , j_m)

$$\bar{i} = \frac{m_{10}}{m_{00}} \qquad \bar{j} = \frac{m_{01}}{m_{00}}$$

(momentos invariantes a la traslación)

$$\mu_{rs} = \sum_{i,j \in \Re} (i - \bar{i})^r (j - \bar{j})^s \qquad \text{for } r,s \in \mathbb{N}$$

Ejemplo: imagen binaria 9x9 - Momentos

j

	1	2	3	4	5	6	7	8	9
1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	1	1	0	0
5	0	0	0	0	1	1	0	0	0
6	0	0	0	1	1	0	0	0	0
7	0	0	1	1	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0

i

En la región R:

$$i = \{4, 4, 5, 5, 6, 6, 7, 7\}$$

$$j = \{3, 4, 4, 5, 5, 6, 6, 7\}$$

$$A = m_{00} = \sum_{i,j \in R} 1 = 8$$

$$m_{10} = \sum_{i \in R} i = 44 \qquad m_{01} = \sum_{j \in R} j = 40$$

$$\bar{i} = \frac{m_{10}}{m_{00}} = 5.5 \qquad \bar{j} = \frac{m_{01}}{m_{00}} = 5$$

$$\mu_{00} = 8, \mu_{10} = 0, \mu_{01} = 0, \mu_{11} = 10, \mu_{20} = 10, \mu_{21} = 20, \mu_{22} = 23$$

Momentos de Hu

Son invariantes al:

- tamaño (escala),
- rotación y
- traslación

$$
\begin{aligned}
\phi_1 &= \eta_{20} + \eta_{02} \\
\phi_2 &= (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2 \\
\phi_3 &= (\eta_{30} - 3\eta_{12})^2 + (3\eta_{21} - \eta_{03})^2 \\
\phi_4 &= (\eta_{30} + \eta_{12})^2 + (\eta_{21} + \eta_{03})^2 \\
\phi_5 &= (\eta_{30} - 3\eta_{12})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] + \\
&\quad (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] \\
\phi_6 &= (\eta_{20} - \eta_{02})[(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] + \\
&\quad 4\eta_{11}(\eta_{30} + \eta_{12})(\eta_{21} + \eta_{03}) \\
\phi_7 &= (3\eta_{21} - \eta_{03})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] - \\
&\quad (\eta_{30} - 3\eta_{12})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2]
\end{aligned}
$$

con

$$\eta_{rs} = \frac{\mu_{rs}}{\mu_{00}^t} \qquad t = \frac{r+s}{2} + 1.$$

Hu, M.K. (1962): Visual Pattern Recognition by moment invariants. IRE Trans. On Information Theory, 8(2): 179-187.

Ejemplo: imagen binaria 9x9 – Momentos de Hu

j

	1	2	3	4	5	6	7	8	9
1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	1	1	0	0
5	0	0	0	0	1	1	0	0	0
6	0	0	0	1	1	0	0	0	0
7	0	0	1	1	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0

i

En la región R:

$$i = \{4, 4, 5, 5, 6, 6, 7, 7\}$$

$$j = \{3, 4, 4, 5, 5, 6, 6, 7\}$$

$$\eta_{20} = \frac{\mu_{20}}{\mu_{00}^2} = 0.1562$$

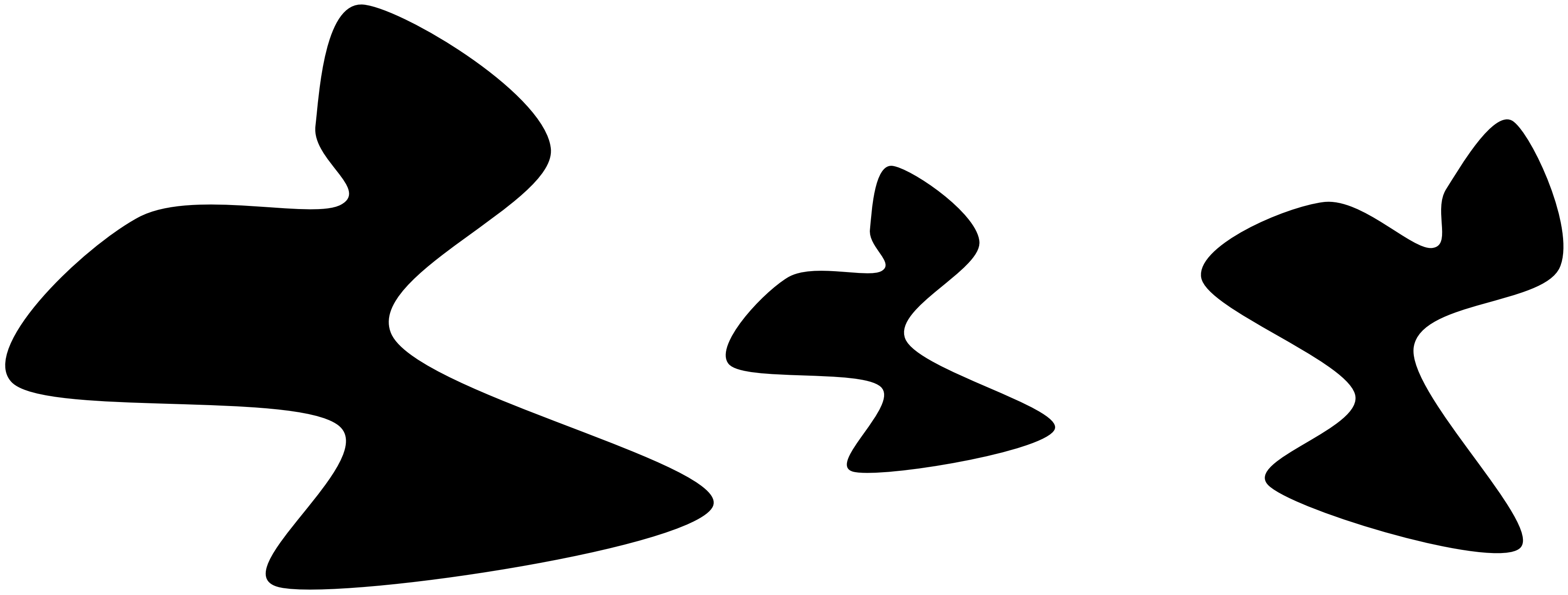
$$\eta_{02} = \frac{\mu_{02}}{\mu_{00}^2} = 0.1875$$

$$\phi_1 = \eta_{20} + \eta_{02} = 0.3438$$

Momentos de Hu

Son invariantes al:

- tamaño (escala),
- rotación y
- traslación



Los valores $\Phi_1, \Phi_2, \ldots \Phi_7$ son similares

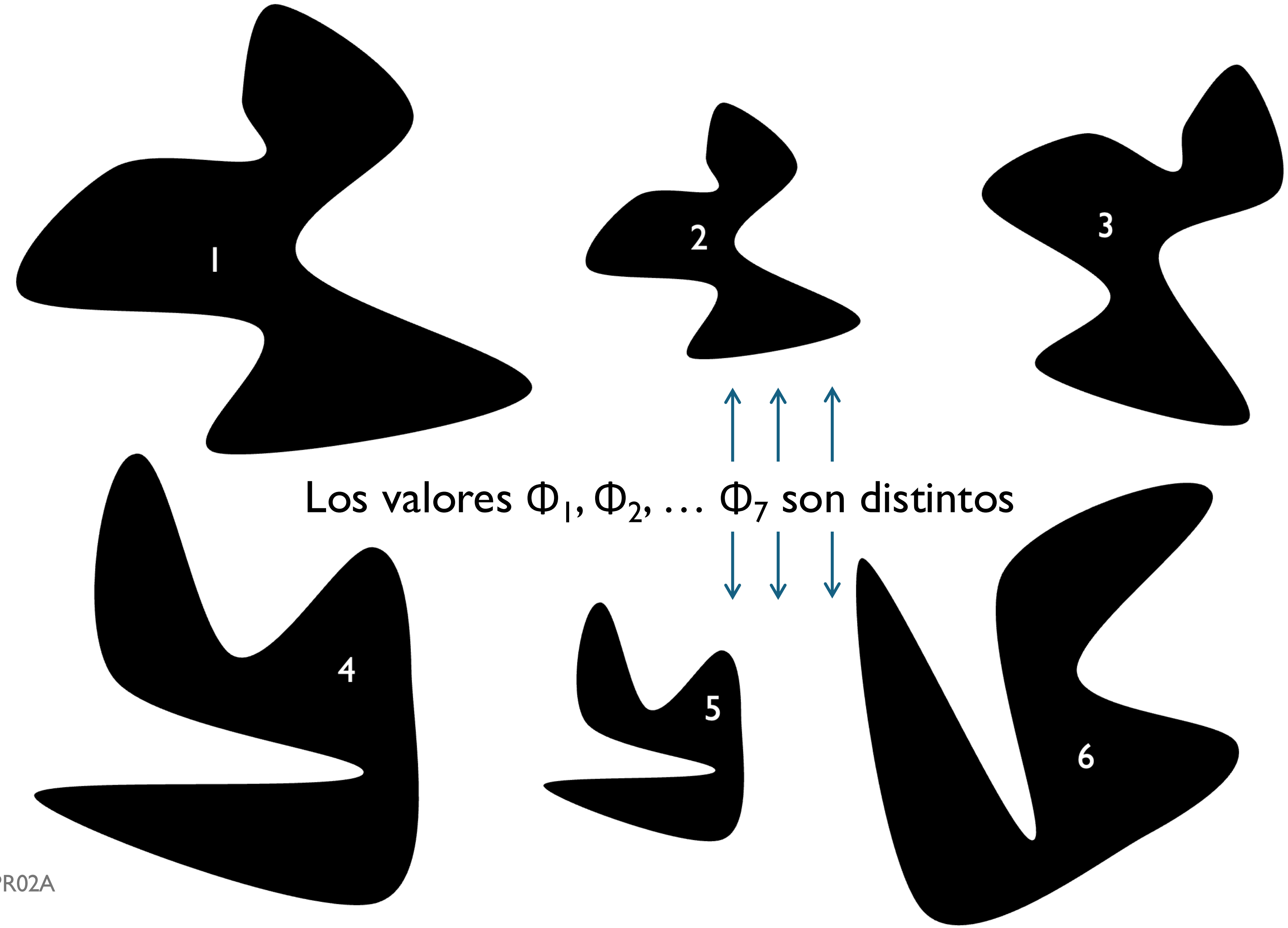
Momentos de Hu

Son invariantes al:

- tamaño (escala),
- rotación y
- traslación

Los valores $\Phi_1, \Phi_2, \ldots \Phi_7$ son similares

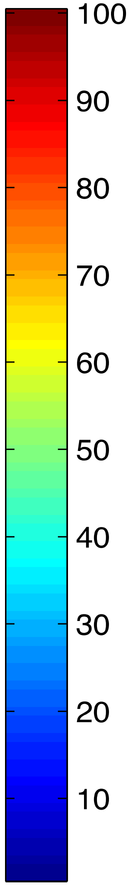
Momentos de Hu

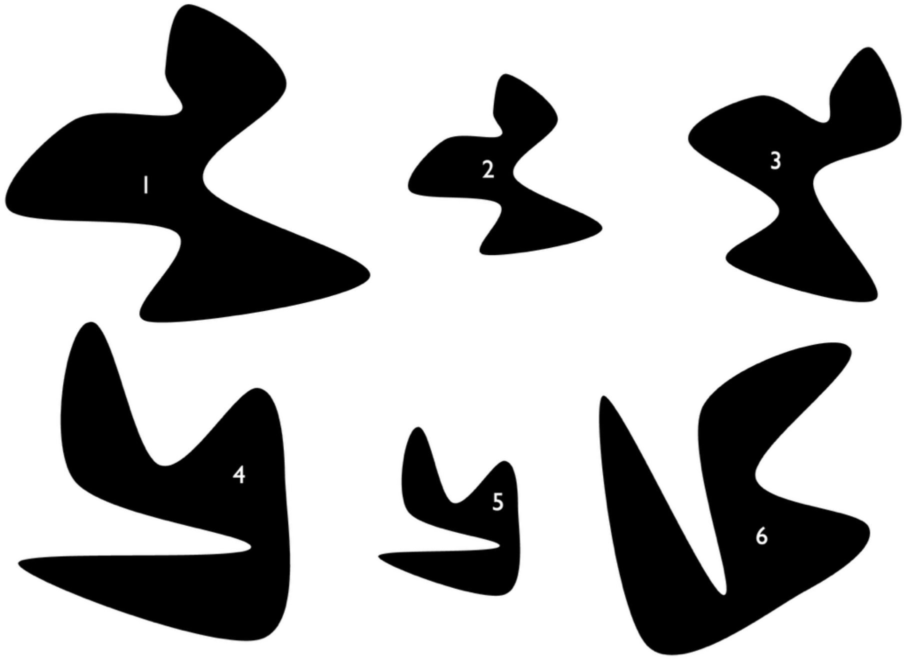


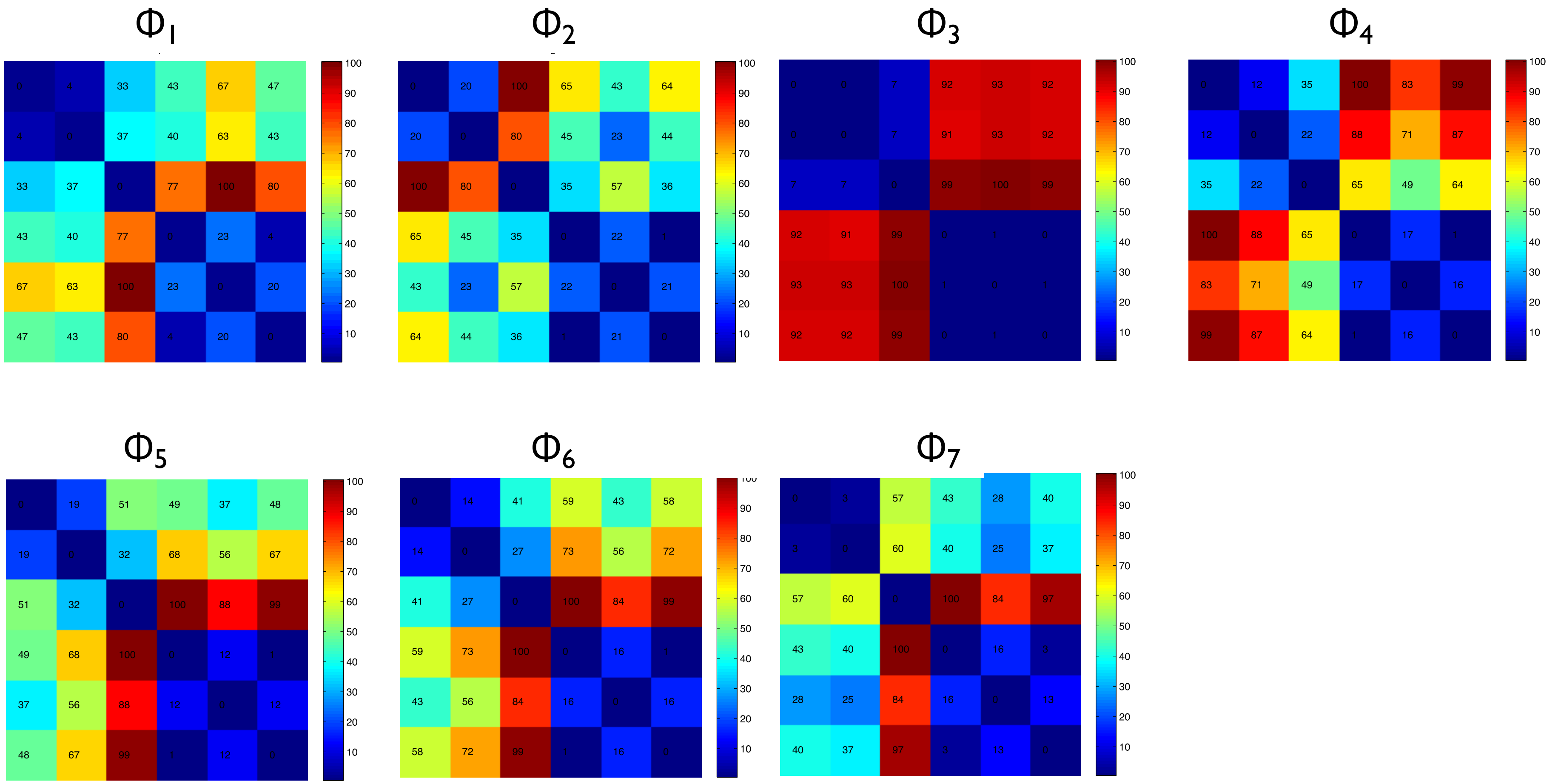
Los valores $\Phi_1, \Phi_2, \ldots \Phi_7$ son distintos

Diferencia entre Φ_1 de la región i y Φ_1 de la región j

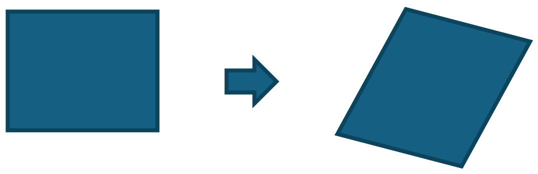
i \ j →	1	2	3	4	5	6
1	0	4	33	43	67	47
2	4	0	37	40	63	43
3	33	37	0	77	100	80
4	43	40	77	0	23	4
5	67	63	100	23	0	20
6	47	43	80	4	20	0






Φ_1
Φ_2
Φ_3
Φ_4
Φ_5
Φ_6
Φ_7

Momentos de Flusser



Invariantes a Escala, Rotación y Traslaciión
+
Invariante a la transformada afín: líneas paralelas se transforman como líneas paralelas

$$I_1 = \frac{\mu_{20}\mu_{02} - \mu_{11}^2}{\mu_{00}^4}$$

$$I_2 = \frac{\mu_{30}^2\mu_{03}^2 - 6\mu_{30}\mu_{21}\mu_{12}\mu_{03} + 4\mu_{30}\mu_{12}^3 + 4\mu_{21}^3\mu_{03} - 3\mu_{21}^2\mu_{12}^2}{\mu_{00}^{10}}$$

$$I_3 = \frac{\mu_{20}(\mu_{21}\mu_{03} - \mu_{12}^2) - \mu_{11}(\mu_{30}\mu_{03} - \mu_{21}\mu_{12}) + \mu_{02}(\mu_{30}\mu_{12} - \mu_{21}^2)}{\mu_{00}^7}$$

$$\begin{aligned} I_4 = \; & (\mu_{20}^3\mu_{03}^2 - 6\mu_{20}^2\mu_{11}\mu_{12}\mu_{03} - 6\mu_{20}^2\mu_{02}\mu_{21}\mu_{03} + 9\mu_{20}^2\mu_{02}\mu_{12}^2 \\ & +12\mu_{20}\mu_{11}^2\mu_{21}\mu_{03} + 6\mu_{20}\mu_{11}\mu_{02}\mu_{30}\mu_{03} - 18\mu_{20}\mu_{11}\mu_{02}\mu_{21}\mu_{12} \\ & -8\mu_{11}^3\mu_{30}\mu_{03} - 6\mu_{20}\mu_{02}^2\mu_{30}\mu_{12} + 9\mu_{20}\mu_{02}^2\mu_{21} \\ & +12\mu_{11}^2\mu_{02}\mu_{30}\mu_{12} - 6\mu_{11}\mu_{02}^2\mu_{30}\mu_{21} + \mu_{02}^3\mu_{30}^2)/\mu_{00}^{11} \end{aligned}$$

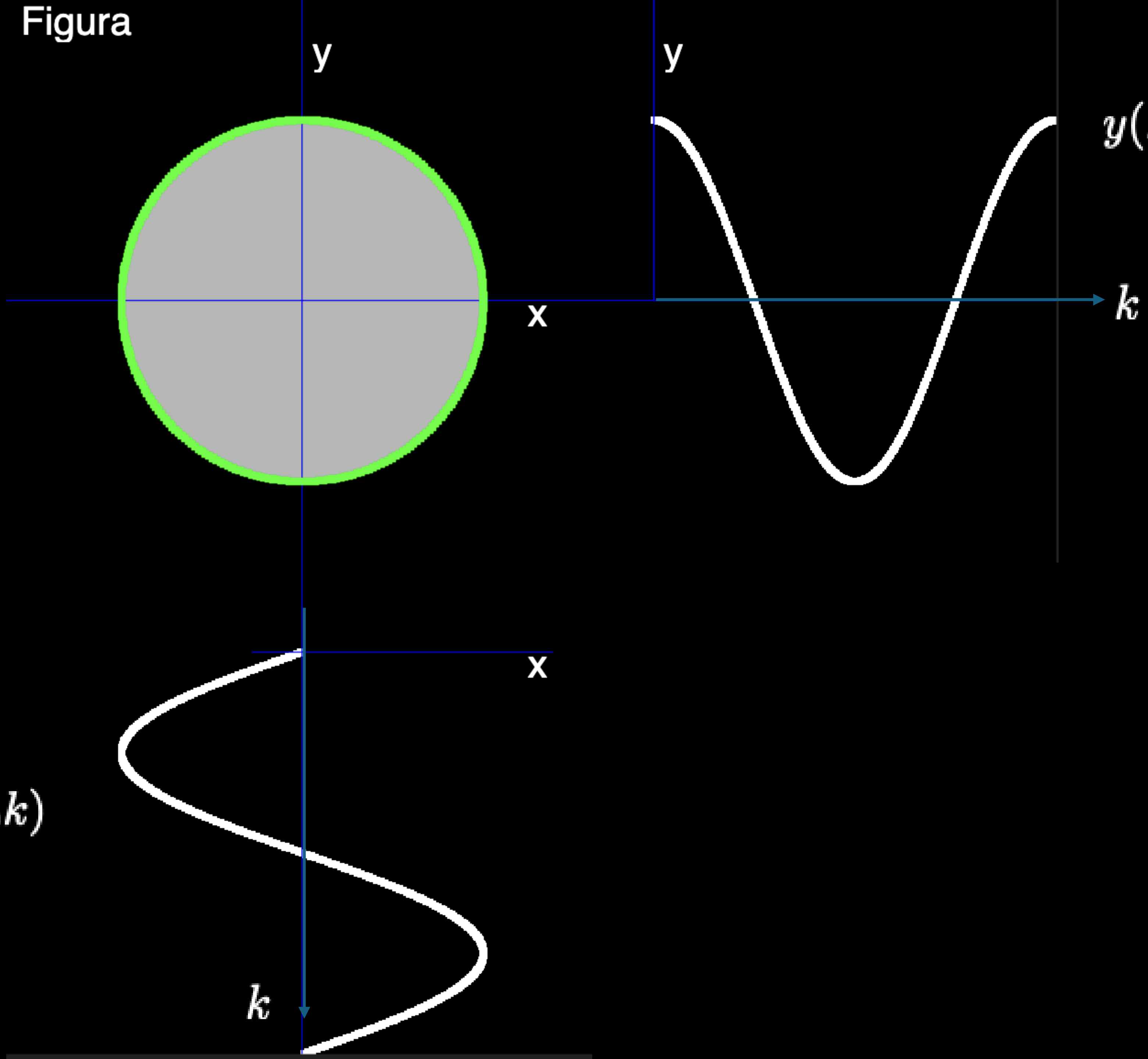
Flusser, J., & Suk, T. (1993). Pattern recognition by affine moment invariants. Pattern recognition, 26(1), 167-174.

Descriptores de Fourier

Descriptores de Fourier

Idea Principal

Figura



$$y(k) = cos(\omega_0 k)$$

$$x(k) = sin(\omega_0 k)$$

$$\omega_0 = \frac{2\pi}{N}$$

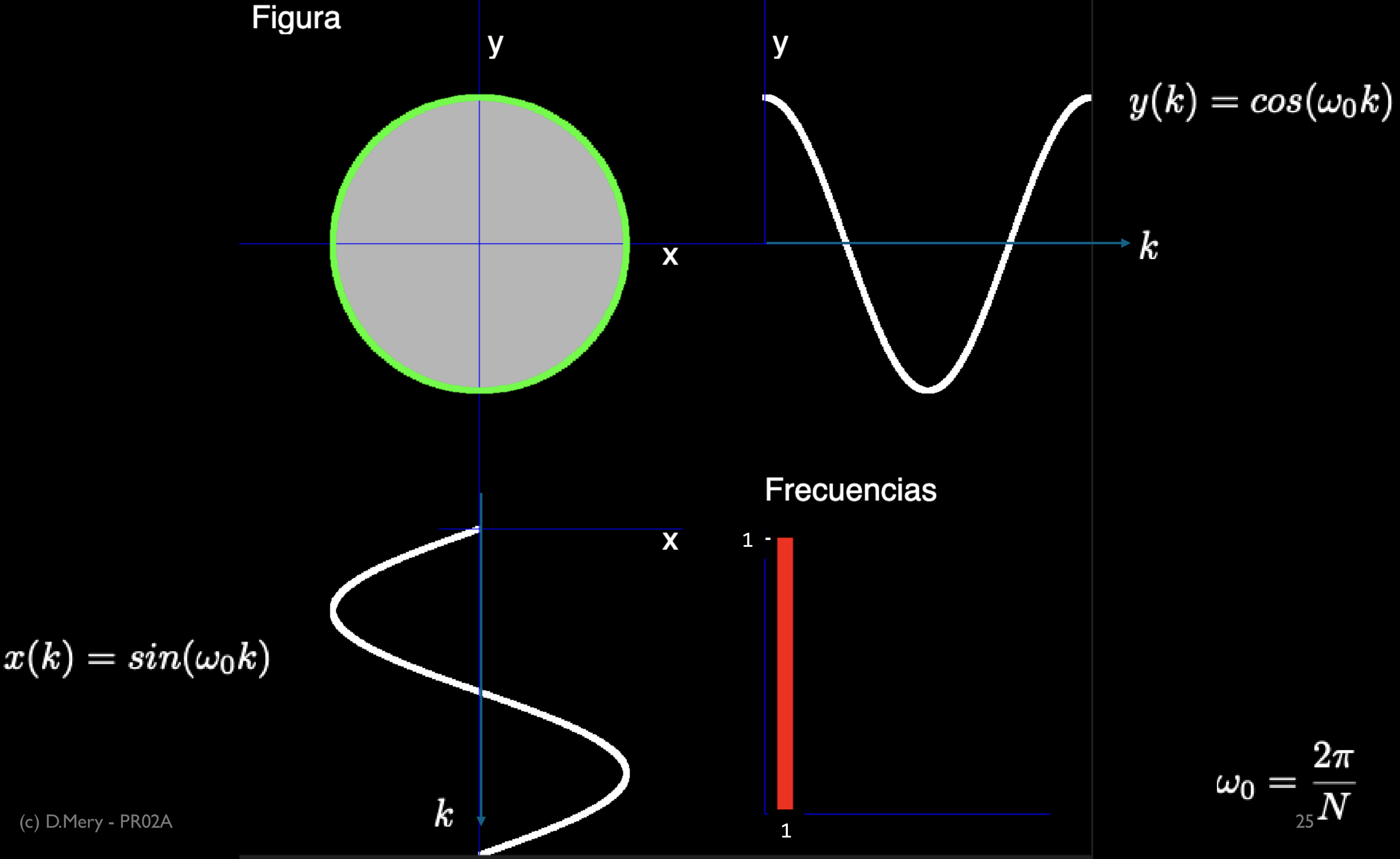
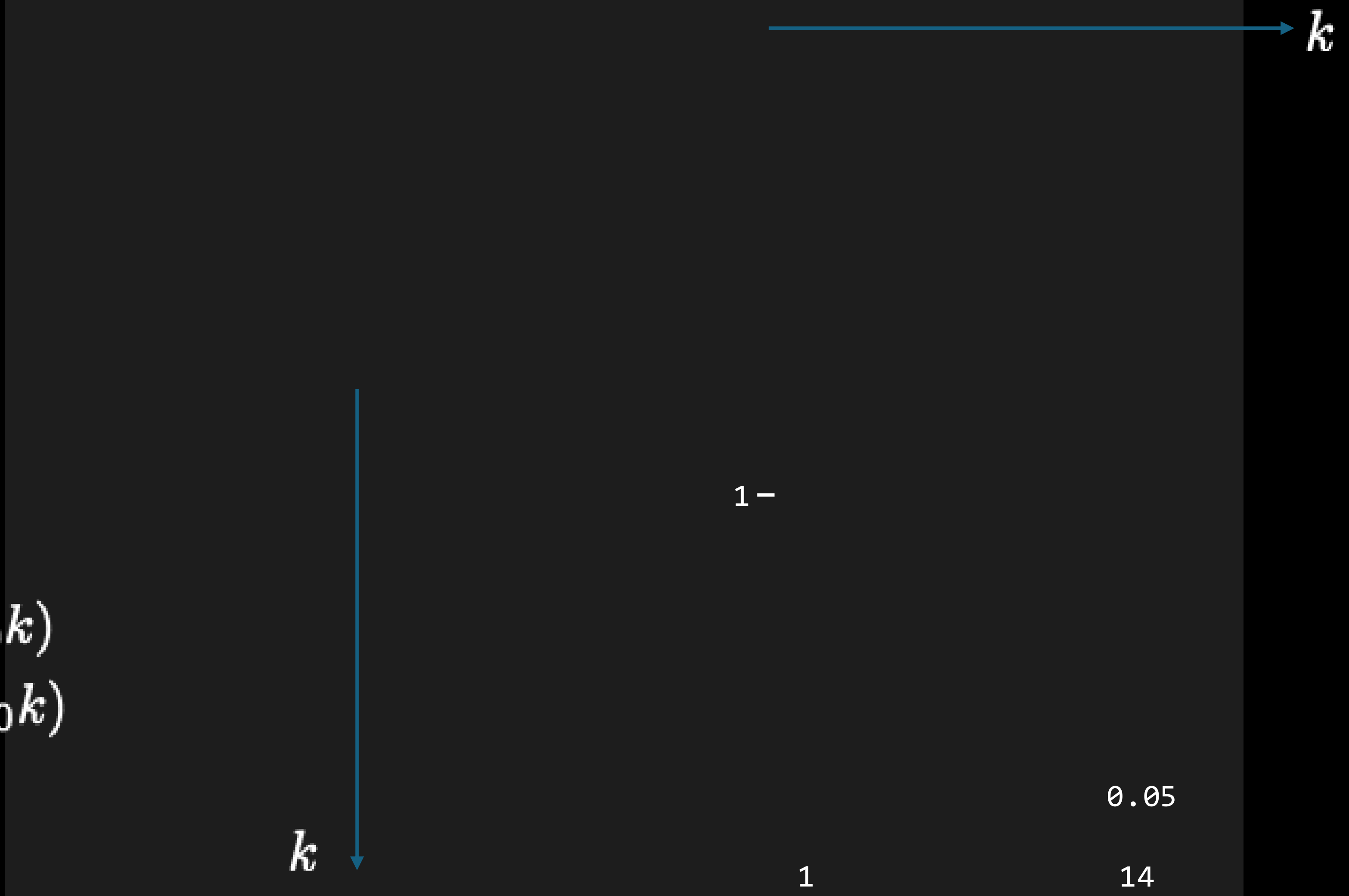


Figura
y
x
y
$y(k) = cos(\omega_0 k)$
k
x
$x(k) = sin(\omega_0 k)$
k
Frecuencias
1
1
$\omega_0 = \frac{2\pi}{N}$

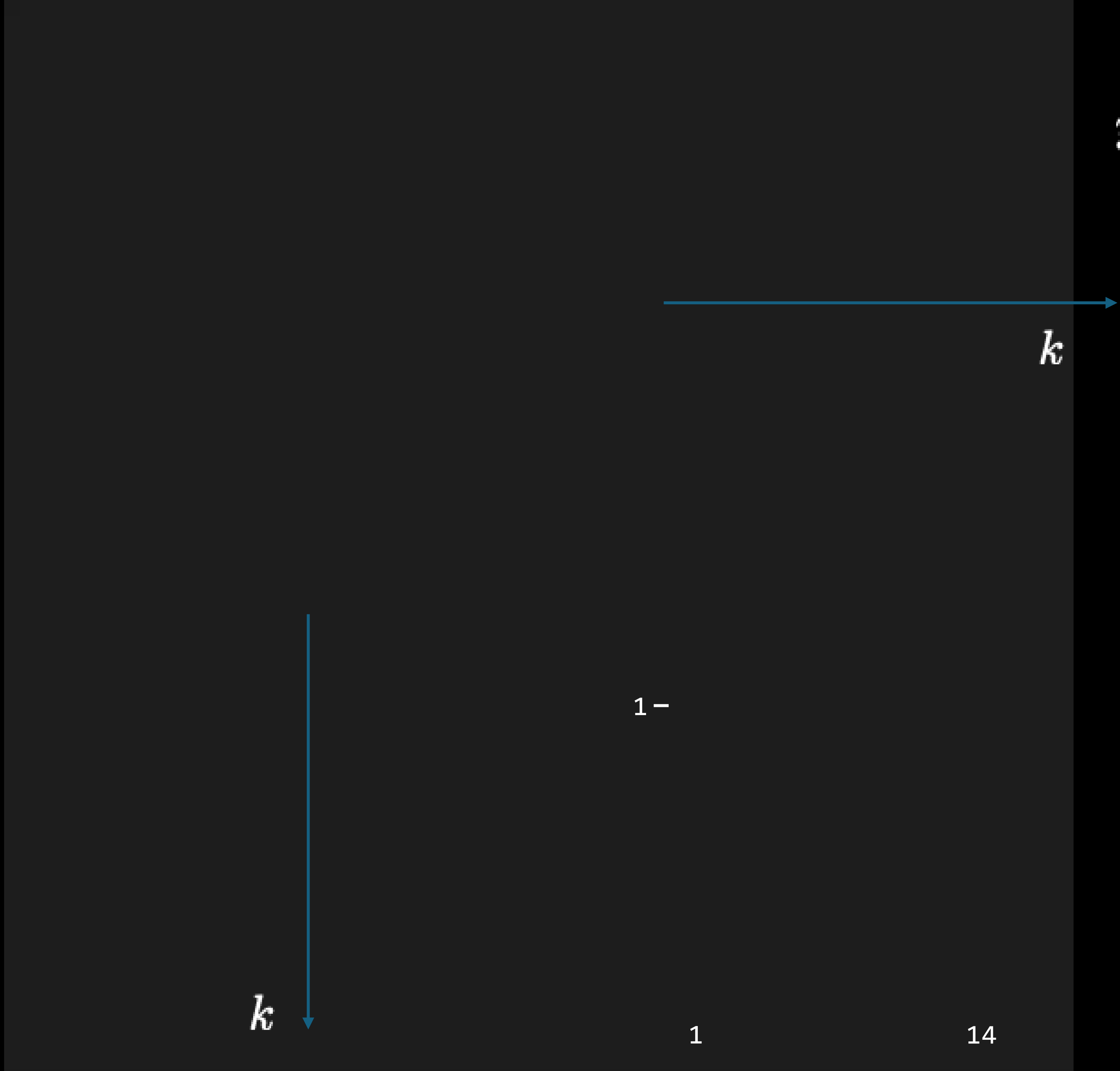
$$y(k) = cos(\omega_0 k)$$
$$+0.05cos(14\omega_0 k)$$



$$x(k) = sin(\omega_0 k)$$
$$+0.05sin(14\omega_0 k)$$

$$\omega_0 = \frac{2\pi}{N}$$

$$x(k) = sin(\omega_0 k)$$
$$+0.02sin(4\omega_0 k)$$
$$+0.04sin(5\omega_0 k)$$
$$+0.07sin(11\omega_0 k)$$
$$+0.05sin(14\omega_0 k)$$



$$y(k) = cos(\omega_0 k)$$
$$+0.02cos(4\omega_0 k)$$
$$+0.04cos(5\omega_0 k)$$
$$+0.07cos(11\omega_0 k)$$
$$+0.05cos(14\omega_0 k)$$

$$\omega_0 = \frac{2\pi}{N}$$

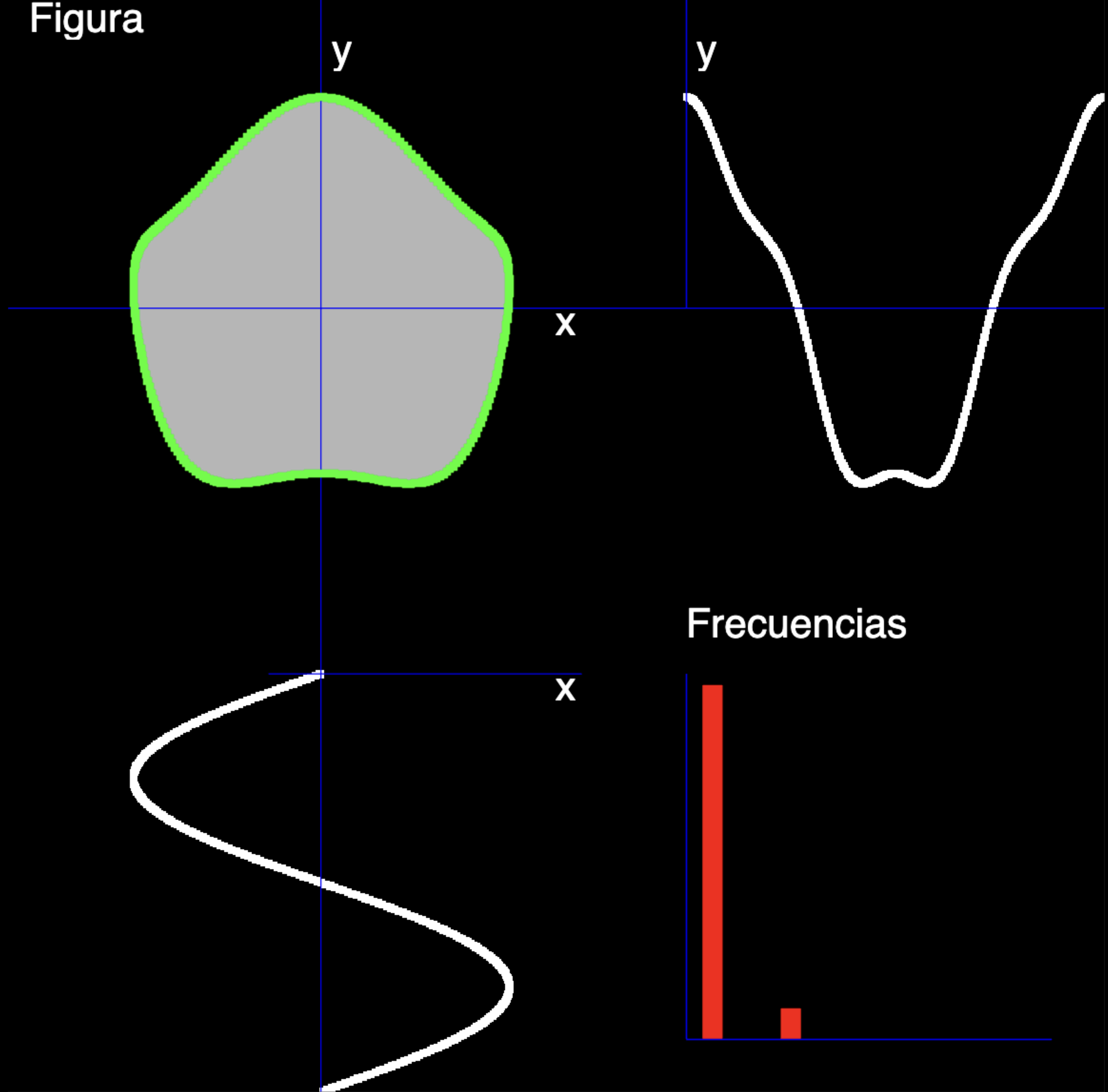


Figura
y
x
y
x
Frecuencias

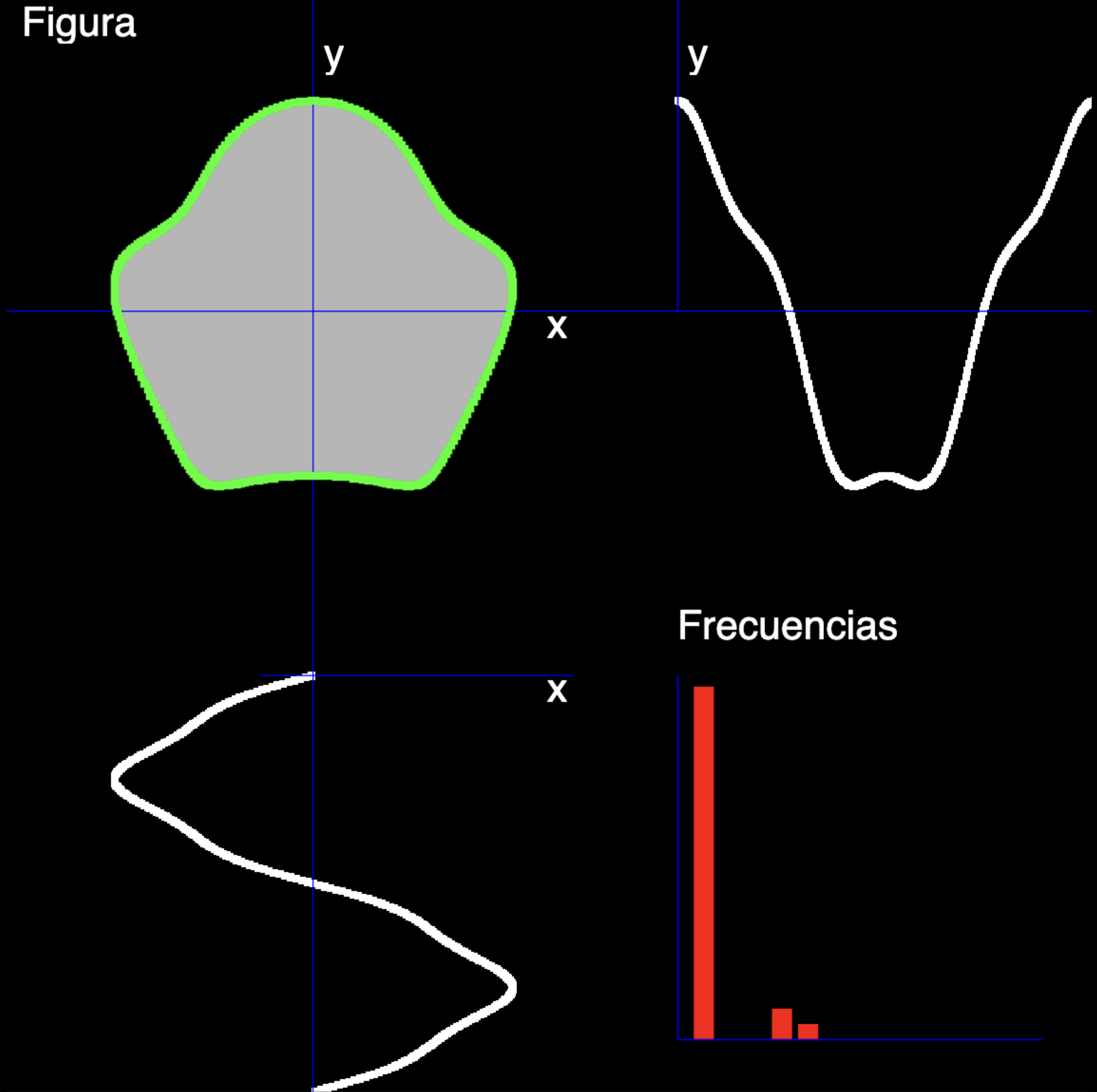


Figura
y
x
y
x
Frecuencias

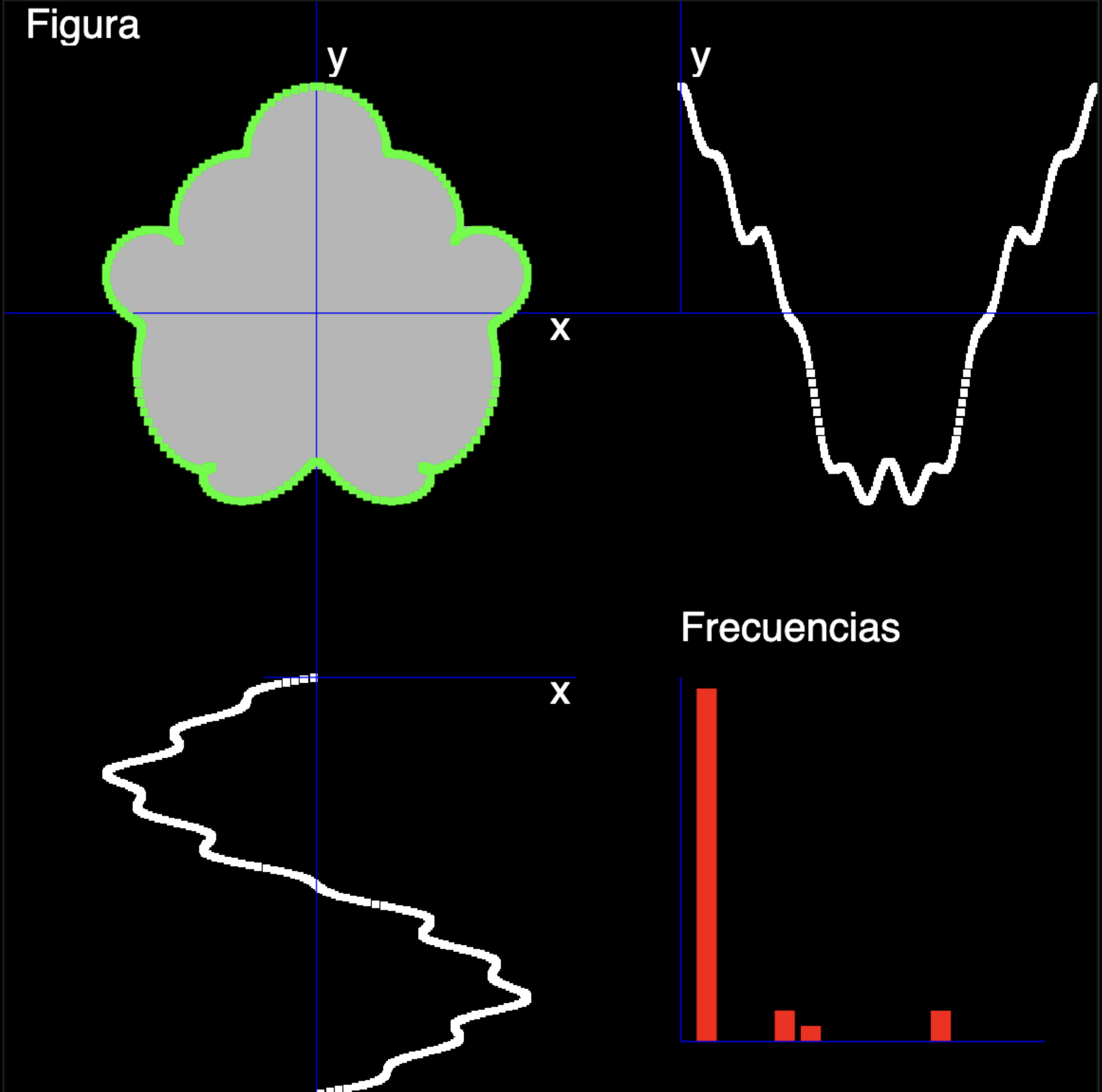


Figura
y
x
y
x
Frecuencias

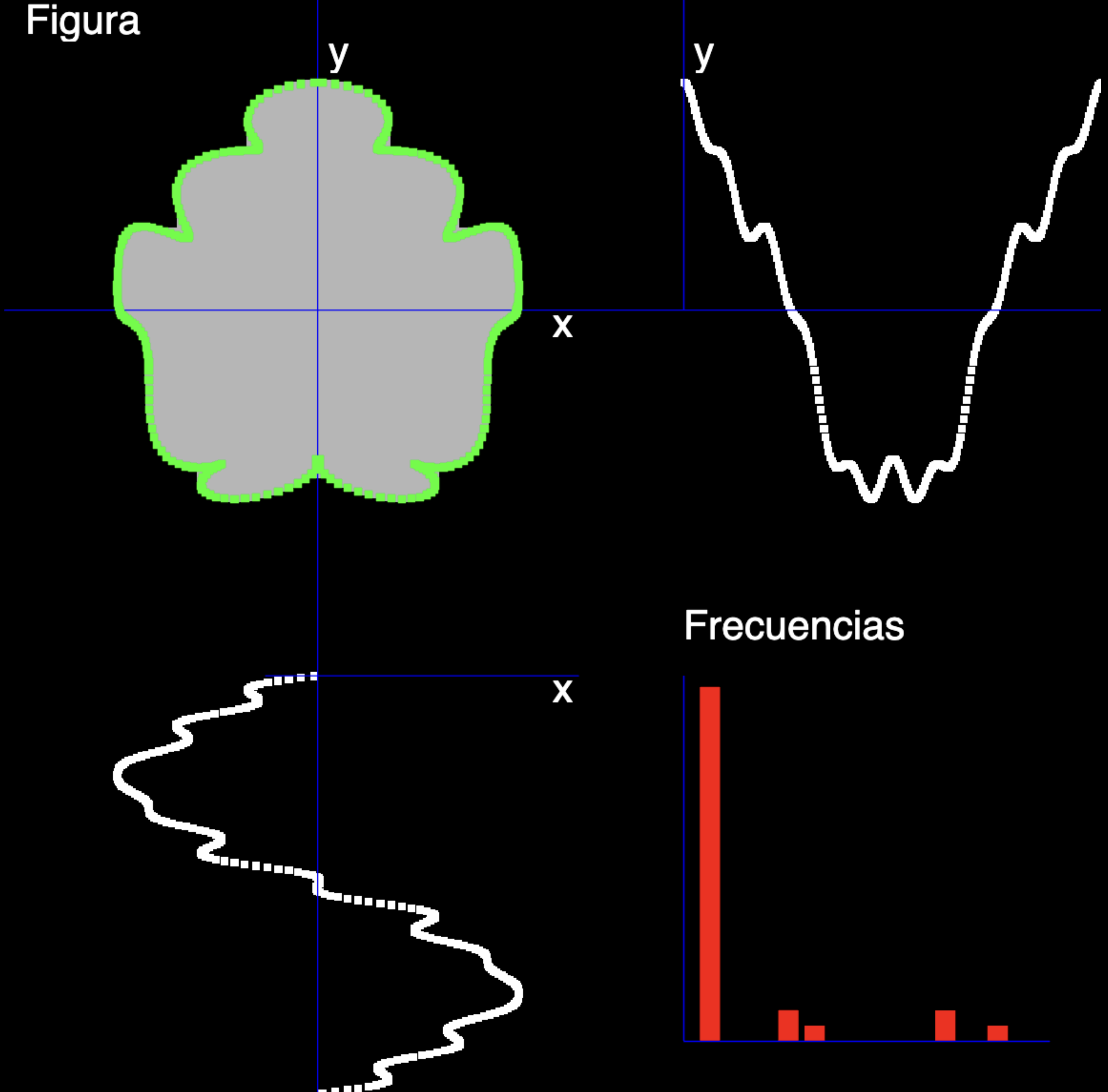


Figura
y
x
y
x
Frecuencias

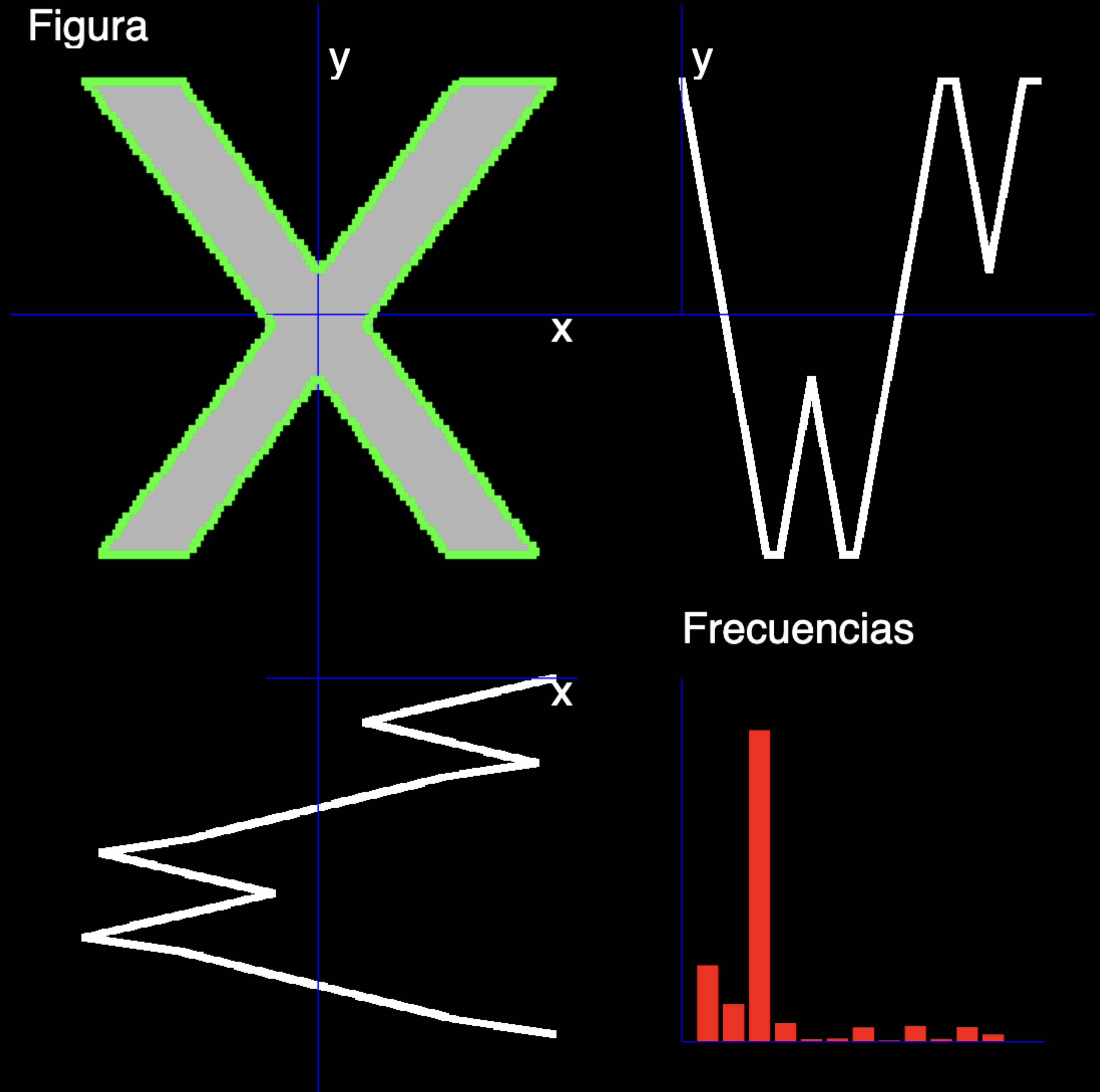
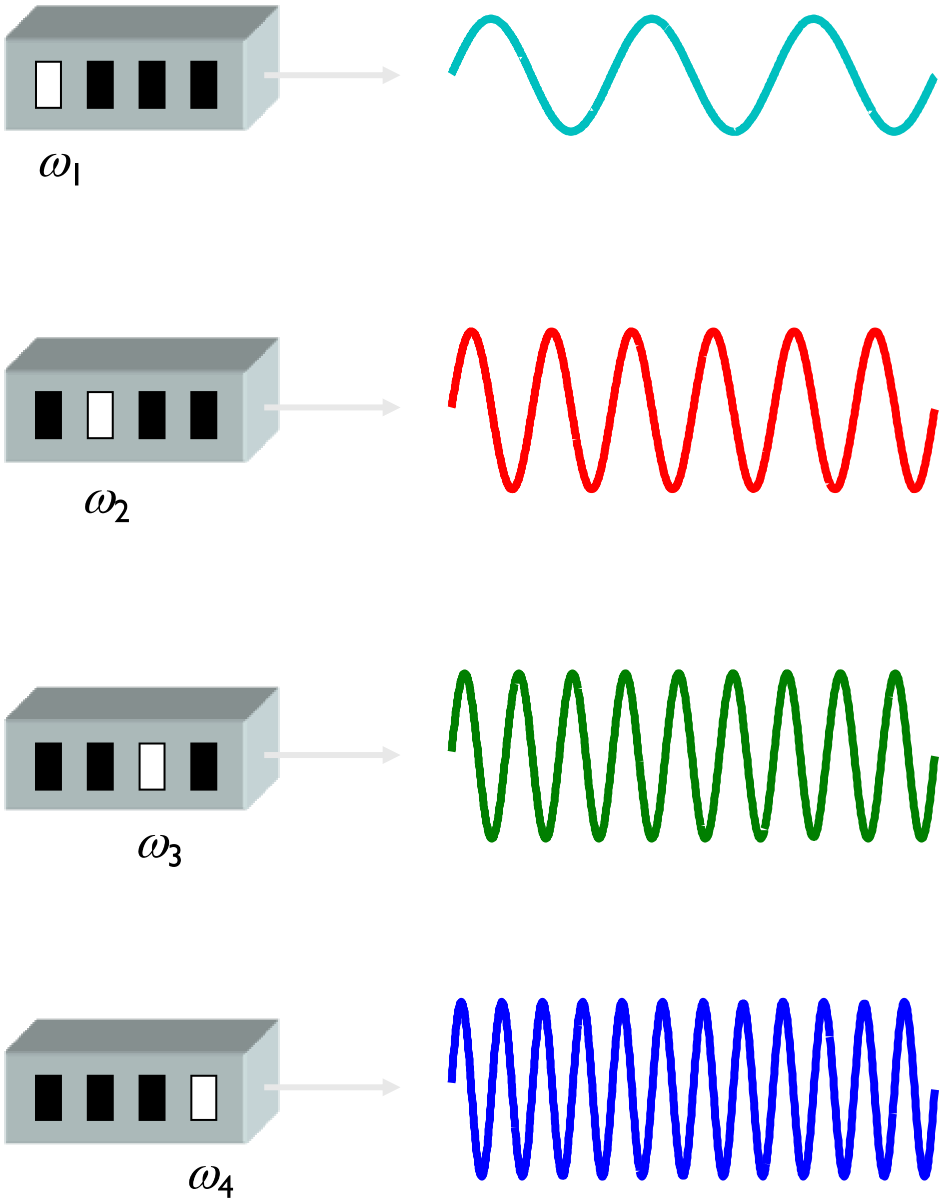


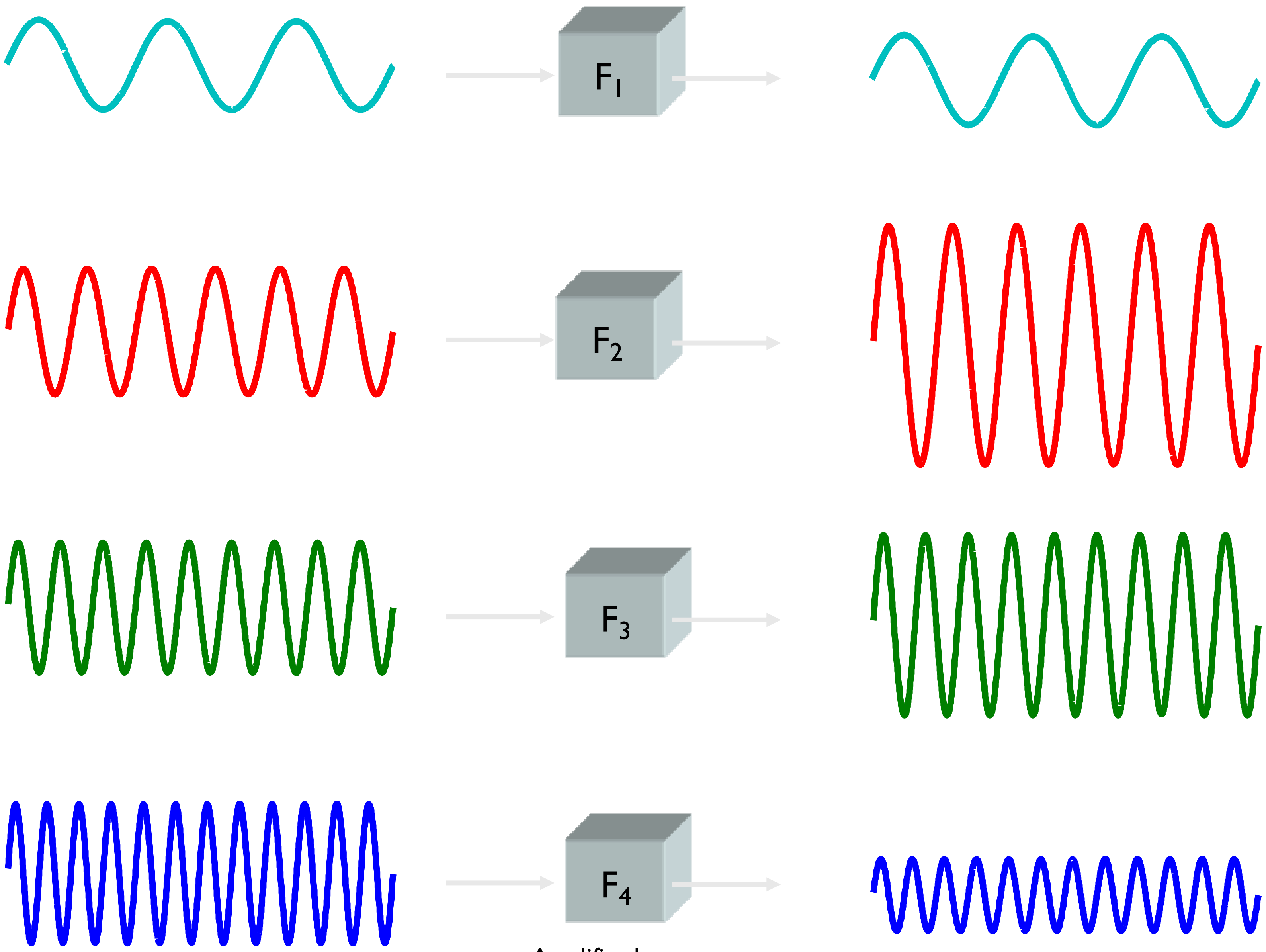
Figura
y
x
y
Frecuencias
x

Descriptores de Fourier

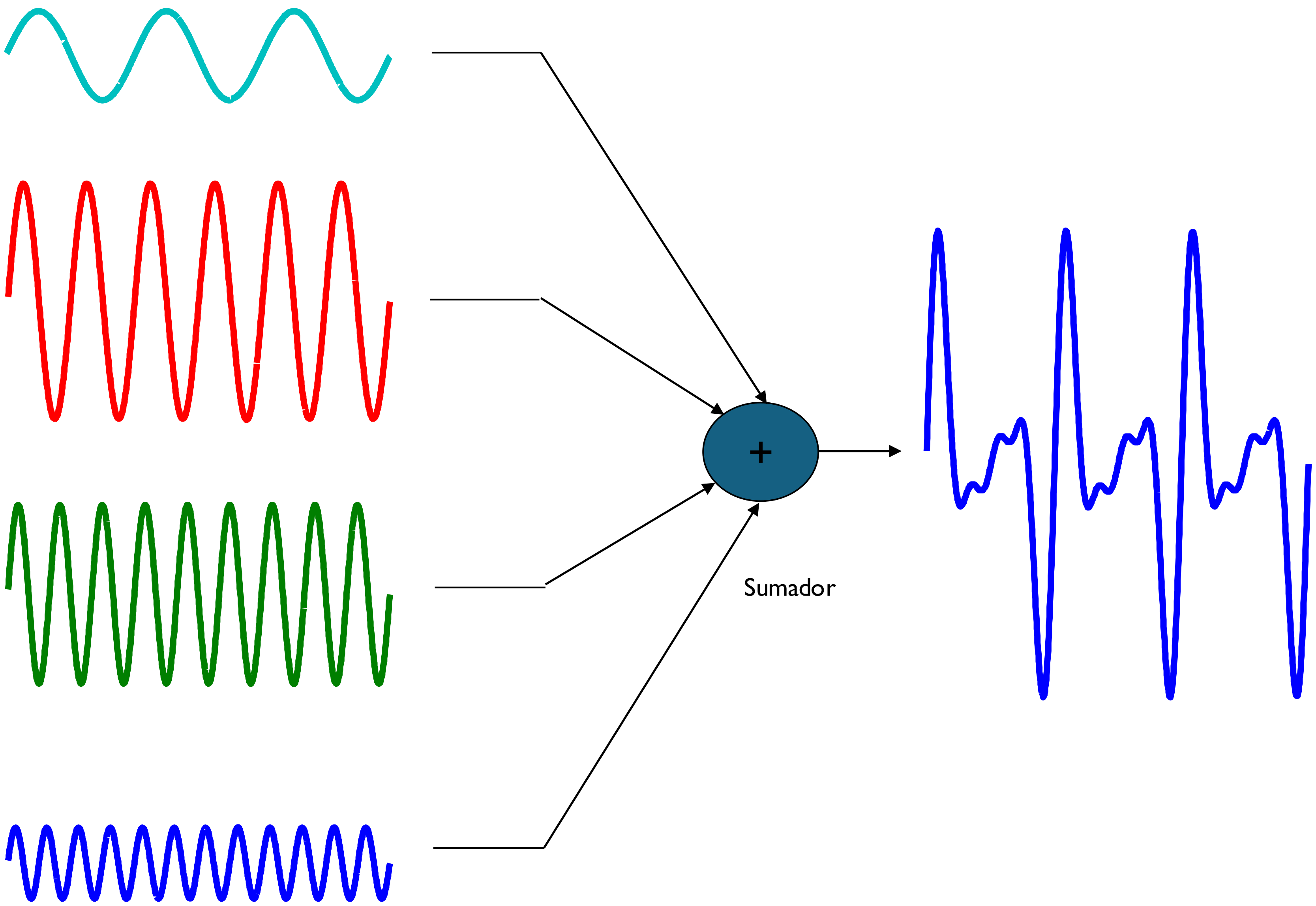
Explicación Simple de la Transformada de Fourier

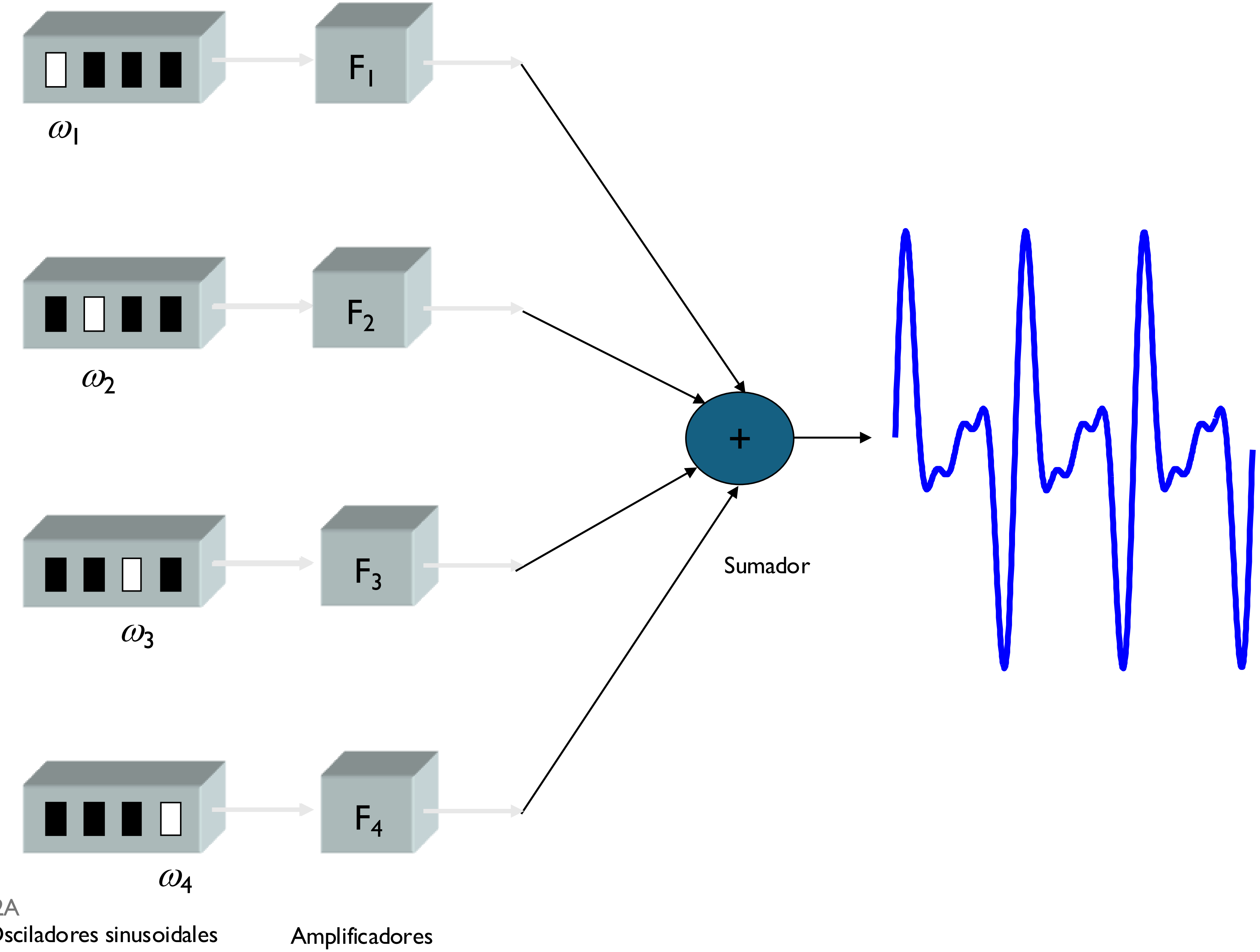


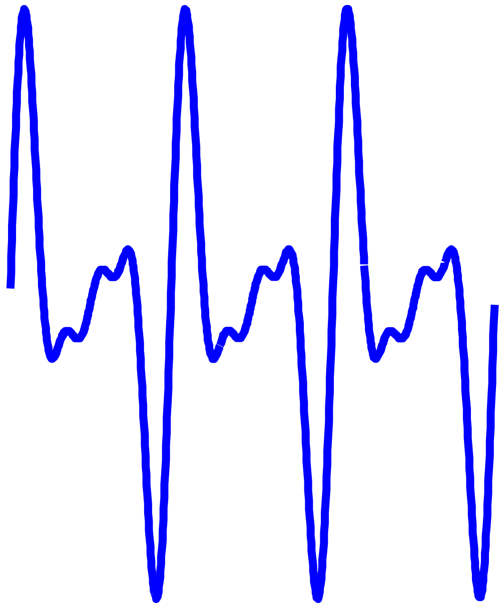
Osciladores sinusoidales



Amplificadores

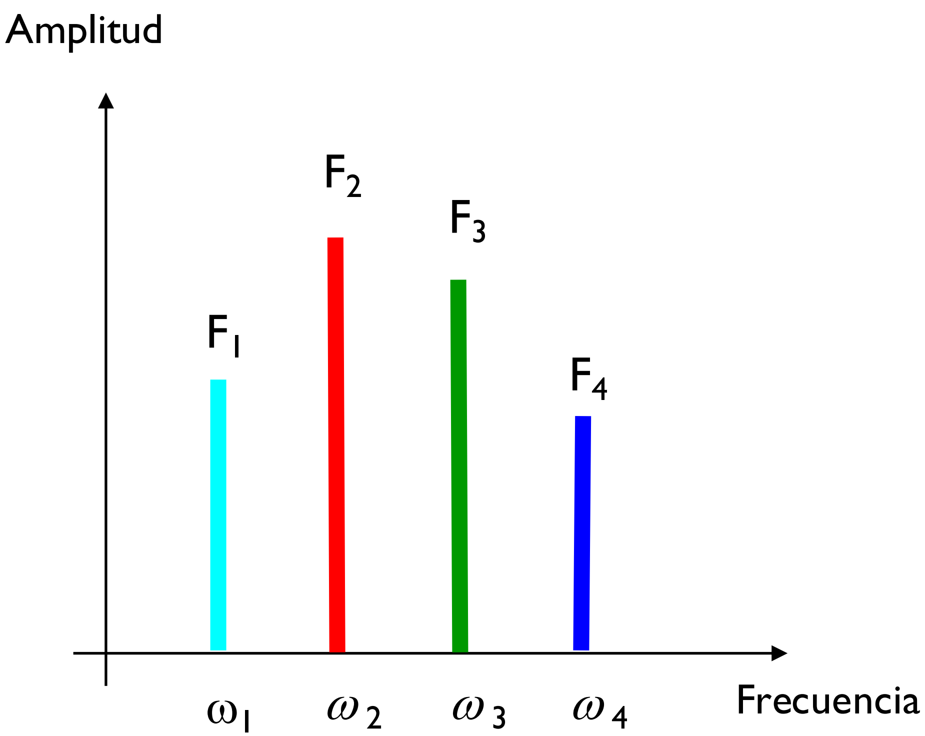

+
Sumador

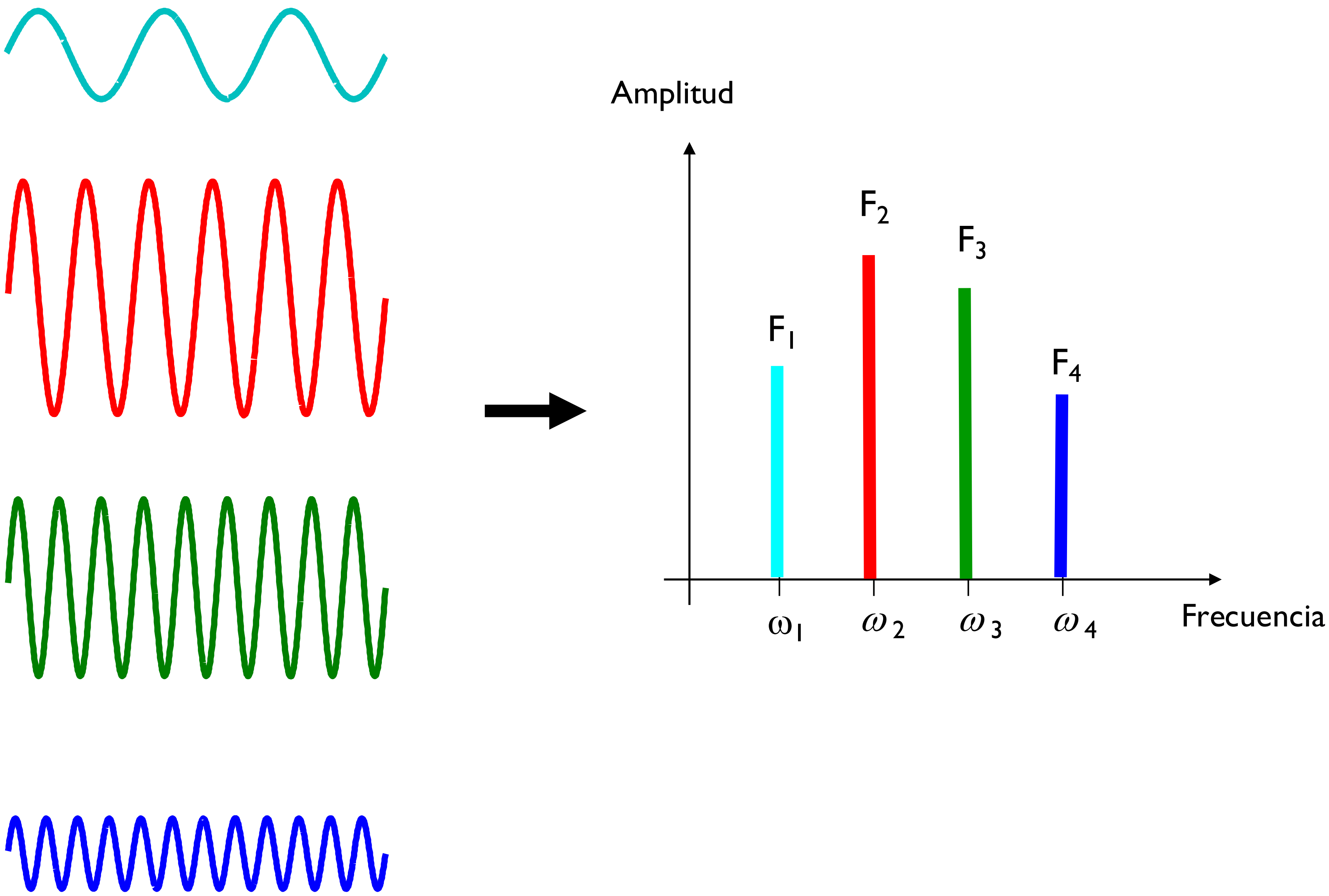

F1
F2
F3
F4
ω_1
ω_2
ω_3
ω_4
+
Sumador
Osciladores sinusoidales
Amplificadores



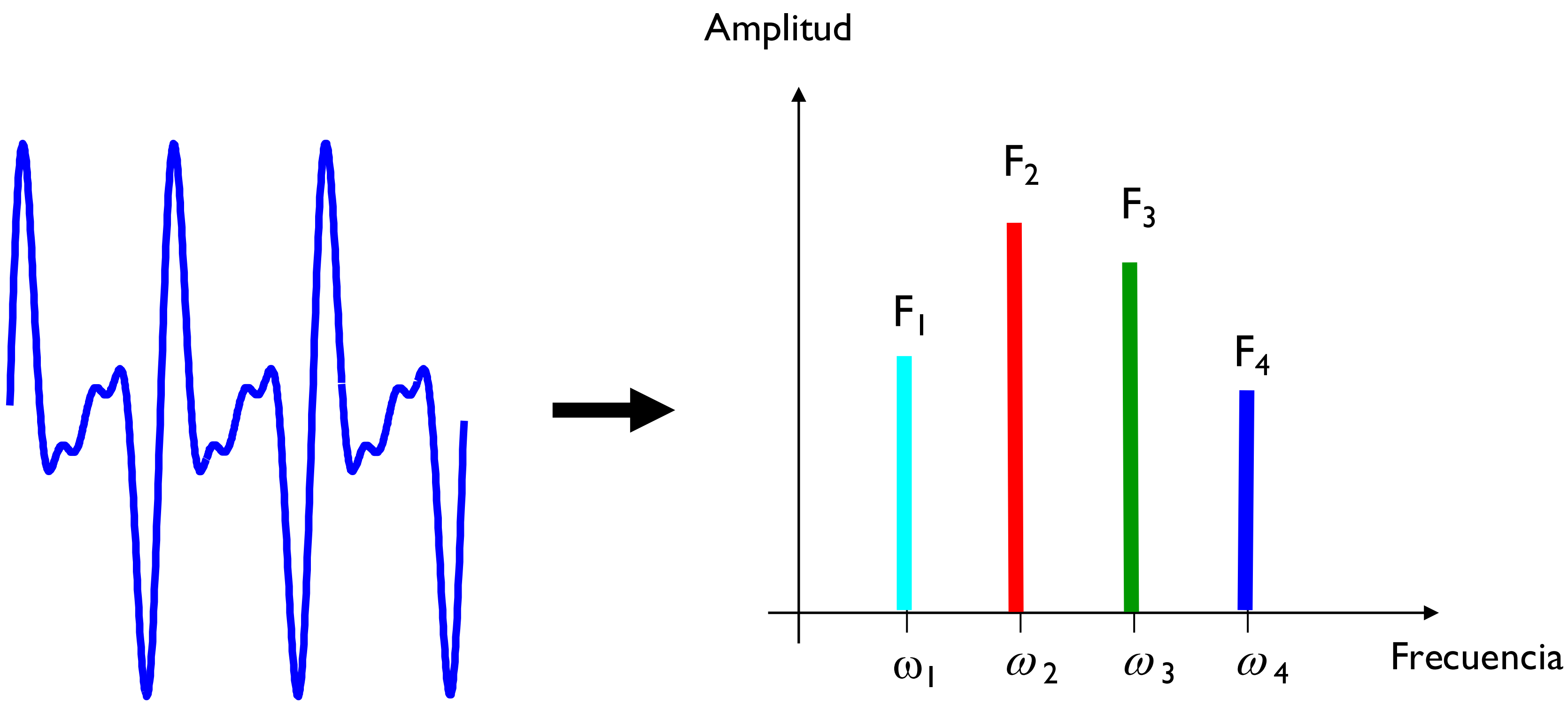
Frecuencia	Amplitud
ω_1	F_1
ω_2	F_2
ω_3	F_3
ω_4	F_4

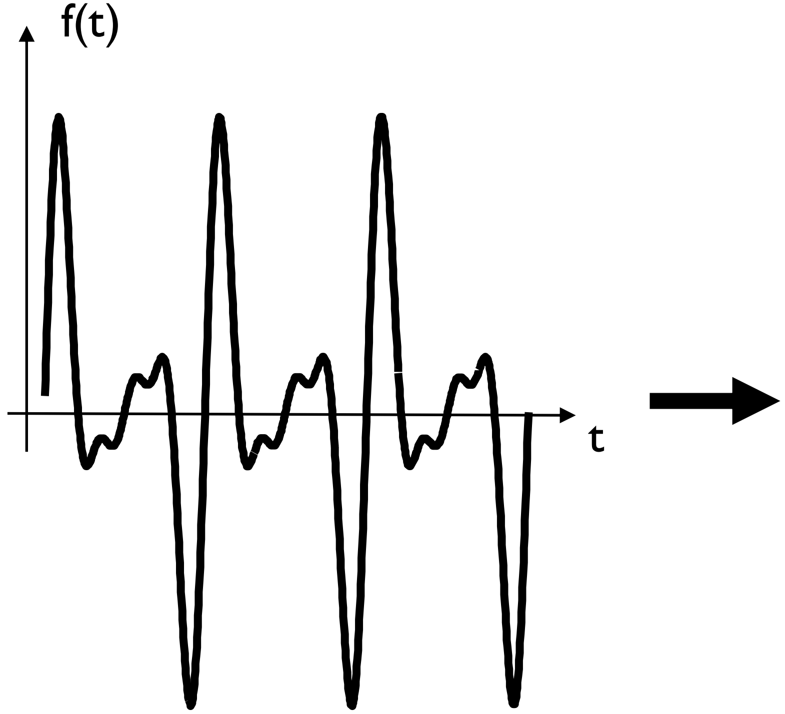
Frecuencia	Amplitud
ω_1	F_1
ω_2	F_2
ω_3	F_3
ω_4	F_4

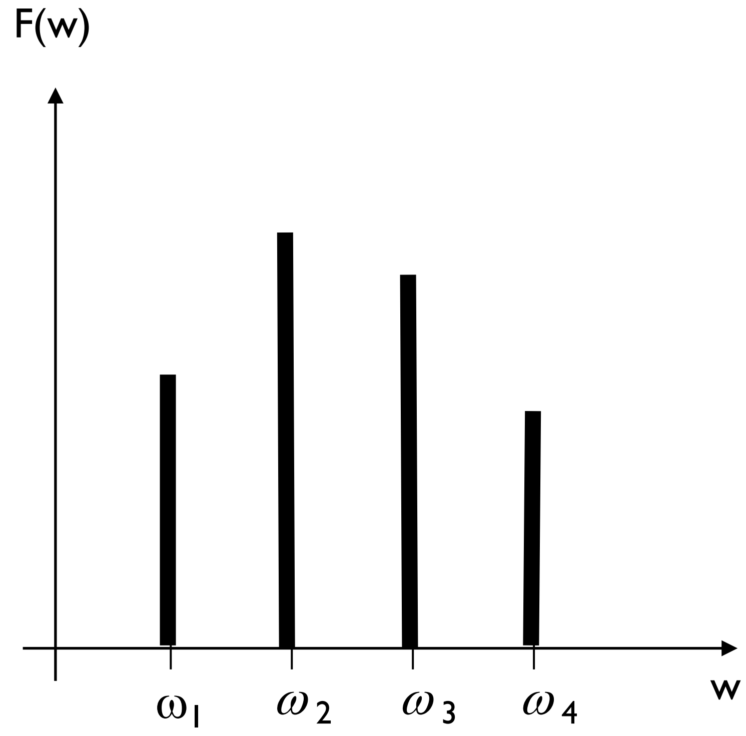




Amplitud
F_1
F_2
F_3
F_4
ω_1
ω_2
ω_3
ω_4
Frecuencia


Amplitud
F_1
F_2
F_3
F_4
ω_1
ω_2
ω_3
ω_4
Frecuencia





Señal f(t)

Espectro de la señal F(w)

f(t) ●—○ F(w)

Descriptores de Fourier

Matemática de la Transformada de Fourier

Transformada de Fourier

CONCEPTOS BÁSICOS

$$f(t) \quad \circ\!\!-\!\!\bullet \quad F(\omega)$$

Función en el dominio del tiempo	Función en el dominio de la frecuencia

Transformada de Fourier

CONTINUA

$$F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-j2\pi\omega t} dt$$

$$f(t) = \int_{-\infty}^{+\infty} F(\omega) e^{+j2\pi\omega t} d\omega$$

$j = \sqrt{-1}$

Transformada de Fourier

DISCRETA

$$F(u) = \frac{1}{N}\sum_{x=0}^{N-1} f(x)e^{-j\frac{2\pi}{N}ux}$$

$$f(x) = \sum_{u=0}^{N-1} F(u)e^{+j\frac{2\pi}{N}ux}$$

$$j = \sqrt{-1}$$

¿Cómo usar Fourier para describir la forma de una figura geométrica?

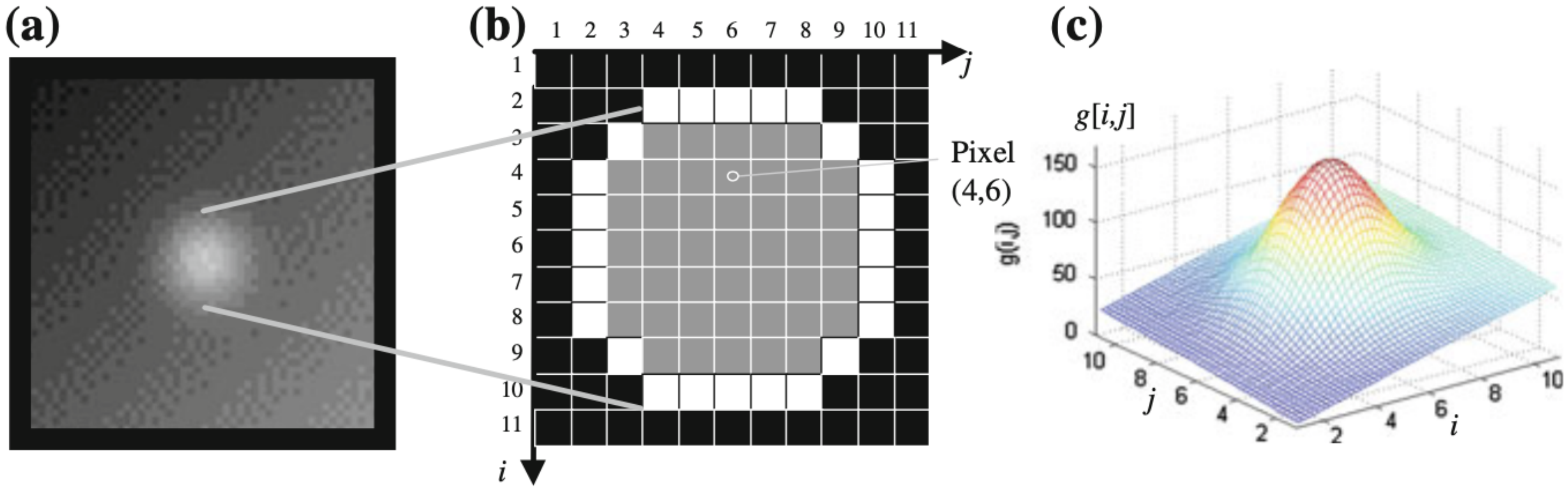


Fig. 5.1 Example of a region: **a** X-ray image, **b** segmented region (*gray* pixels), **c** 3D representation of the gray values

5.2.3 *Fourier Descriptors*

Shape information—invariant to scale, orientation and position—can be measured using *Fourier descriptors* [5–7]. The coordinates of the pixels of the boundary are arranged as a complex number $i_k + j \cdot j_k$, with $j = \sqrt{-1}$ and $k = 0, \ldots, L-1$, where L is the perimeter of the region, and pixel k and $k+1$ are connected. The complex boundary function can be considered as a periodical signal of period L. The Discrete Fourier Transformation [8] gives a characterization of the shape of the region. The Fourier coefficients are defined by:

$$F_n = \sum_{k=0}^{L-1} (i_k + j \cdot j_k) e^{-j\frac{2\pi kn}{L}} \qquad \text{for } n = 0, \ldots, L-1. \tag{5.10}$$

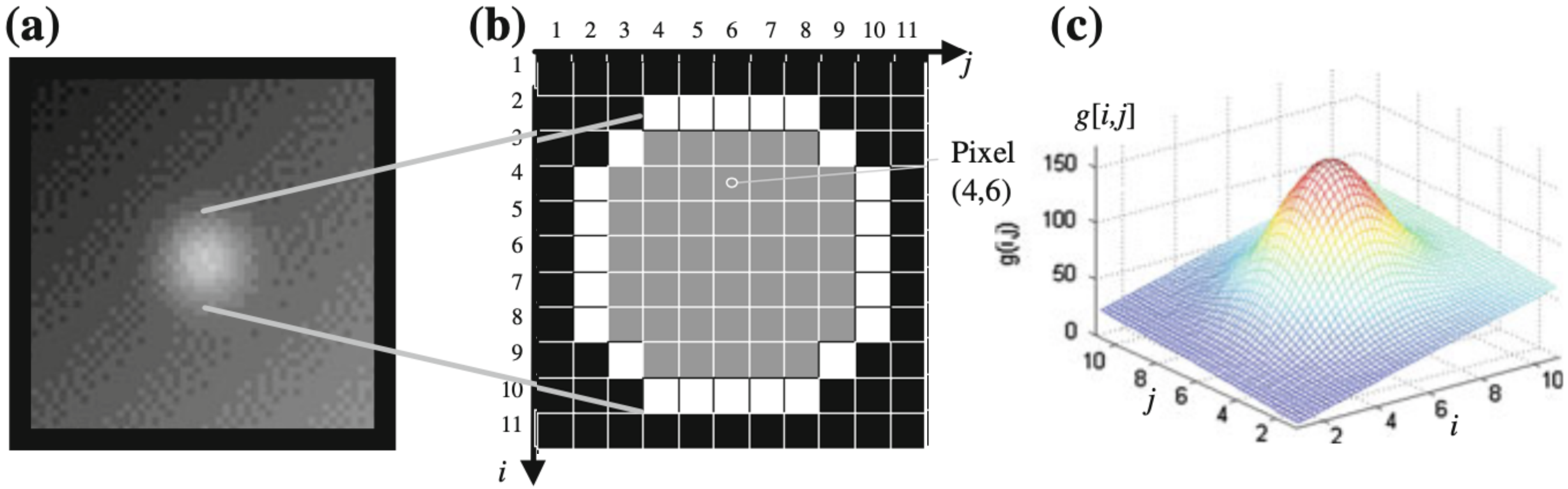
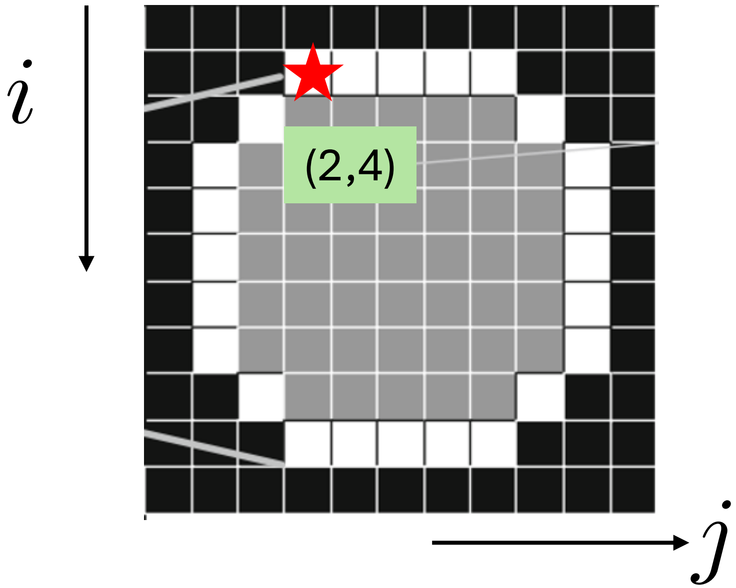
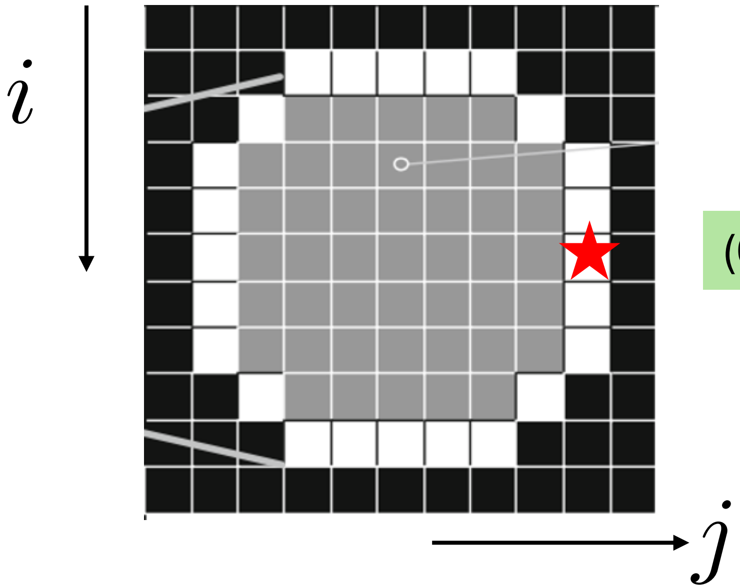


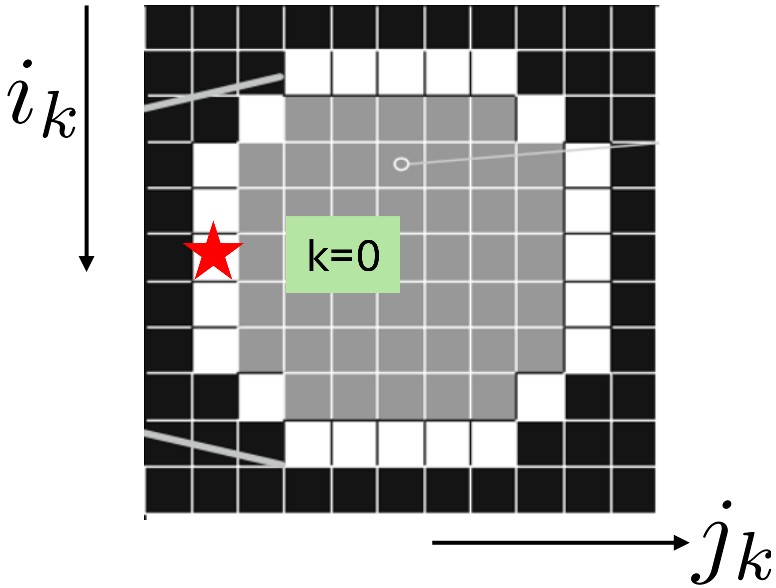
Fig. 5.1 Example of a region: **a** X-ray image, **b** segmented region (*gray* pixels), **c** 3D representation of the gray values

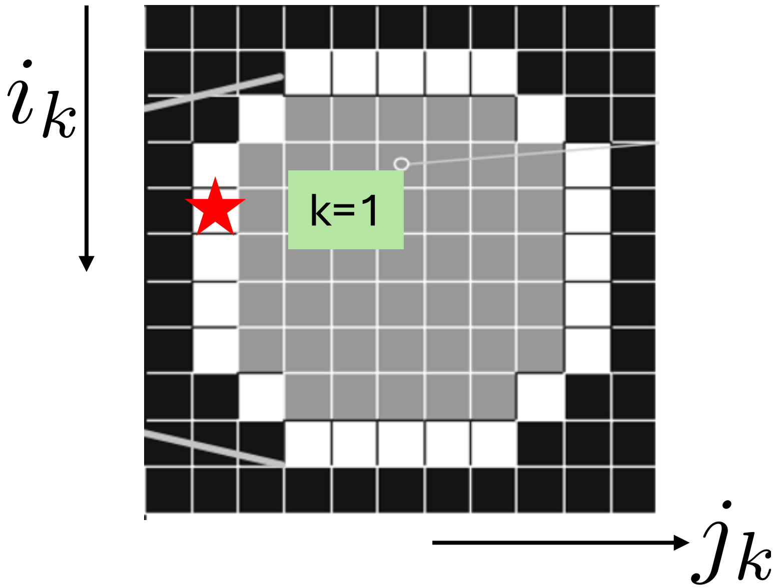


i
(2,4)
j



(6,10)


i_k
k=0
j_k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k=1
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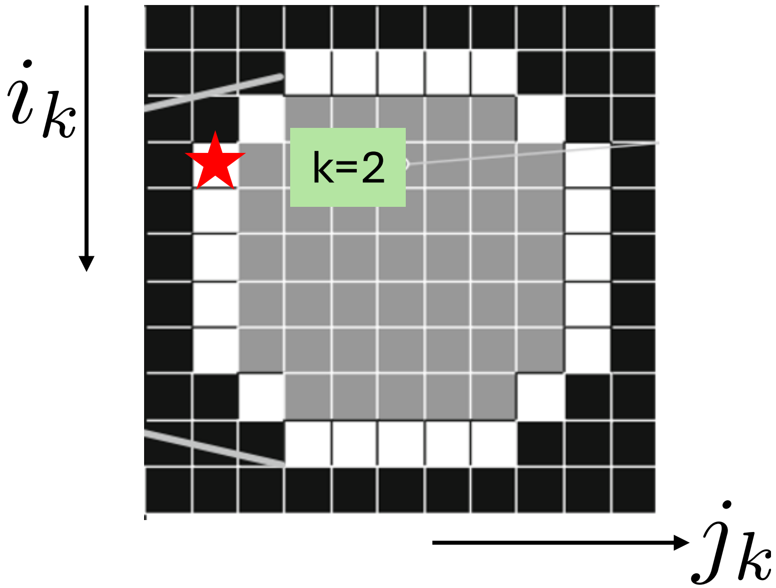
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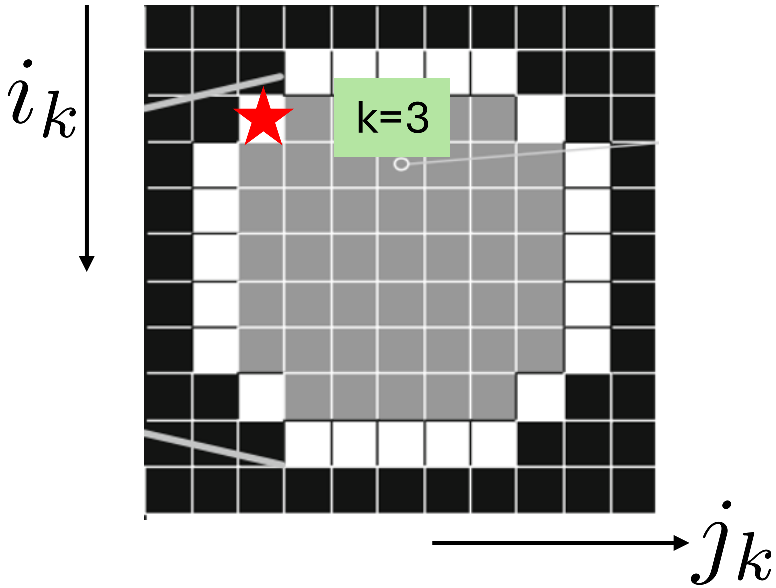
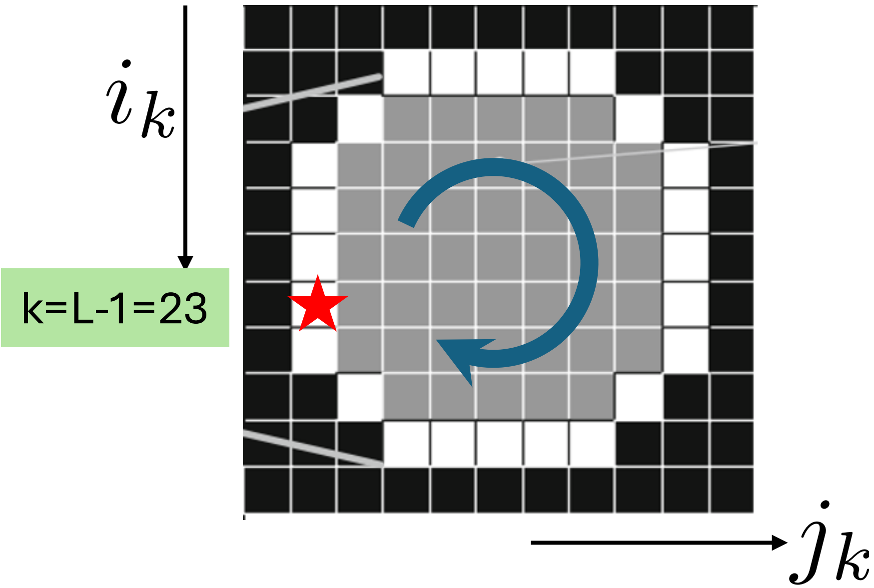
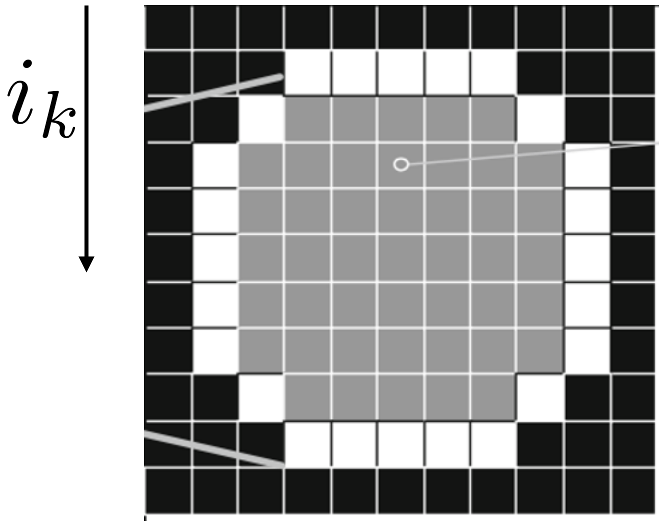
k=2
j_k


k=3



i_k
k=L-1=23
j_k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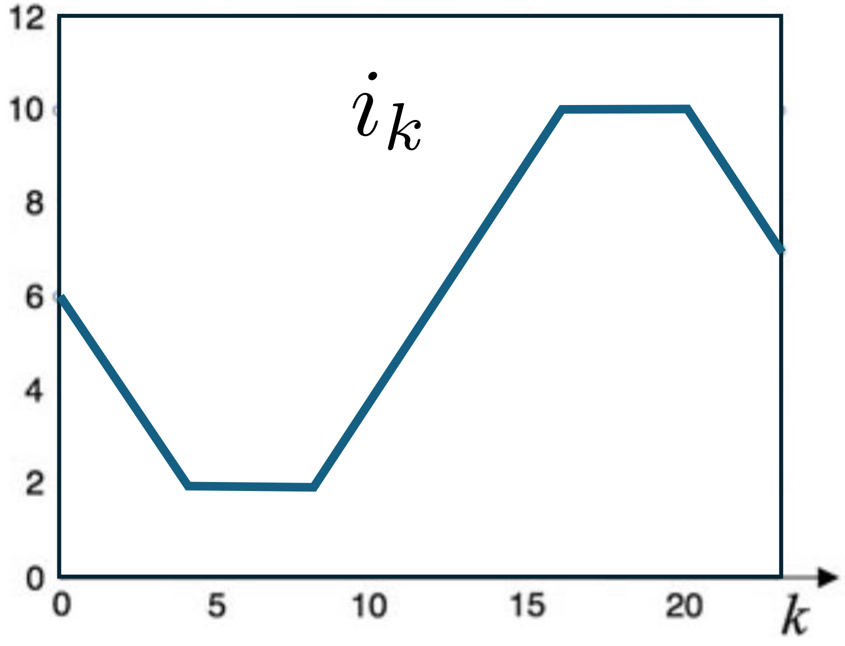
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10
8
6
4
2
0
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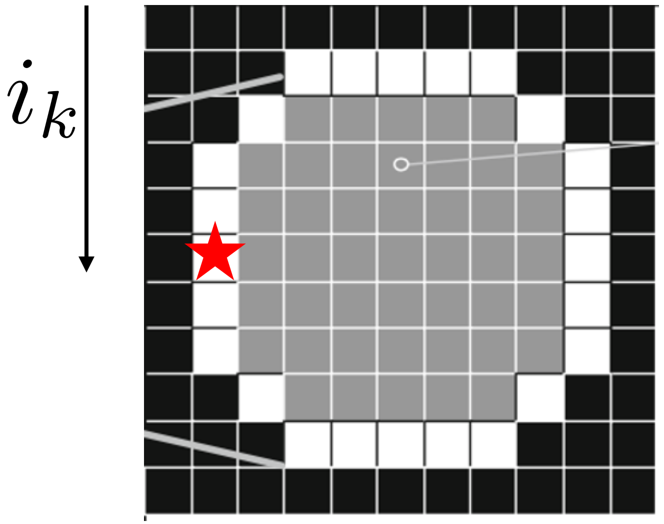
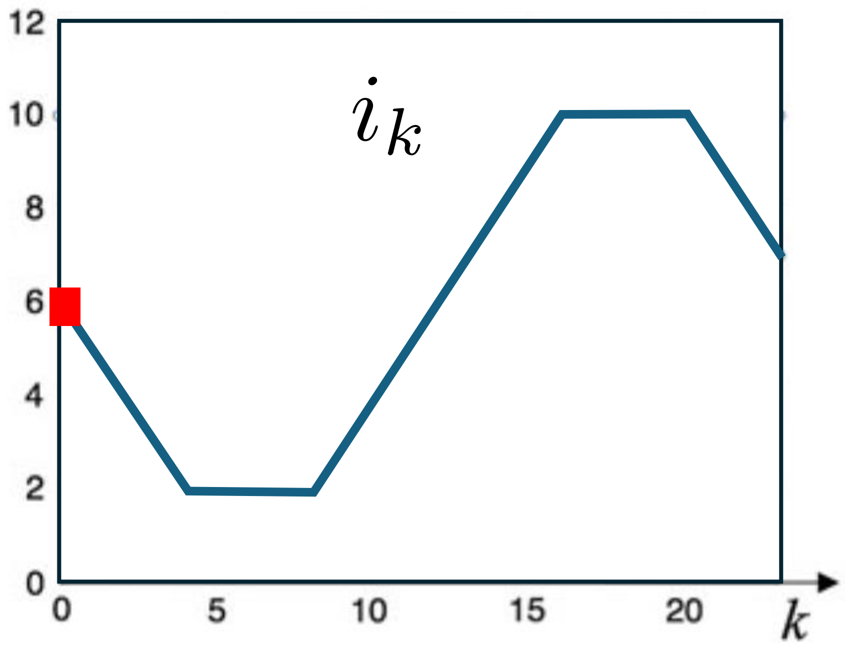
5
10
15
20
i_k
k


i_k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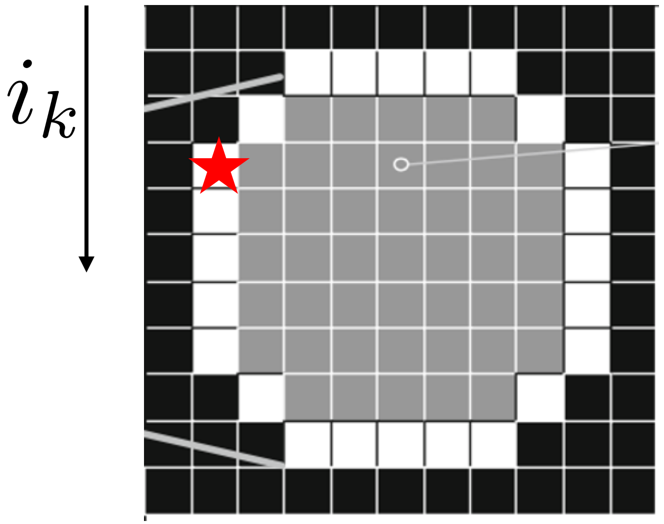
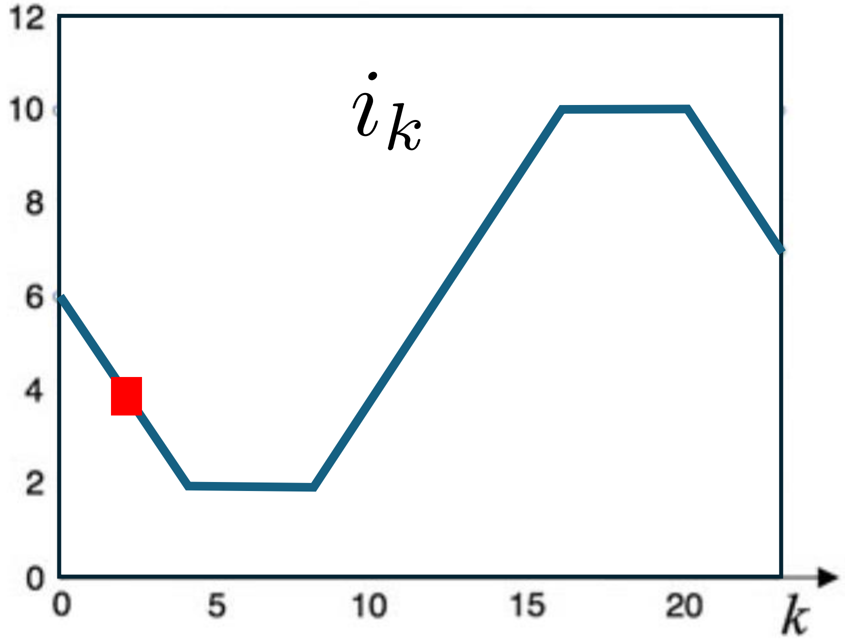
5
10
15
20
k
i_k


i_k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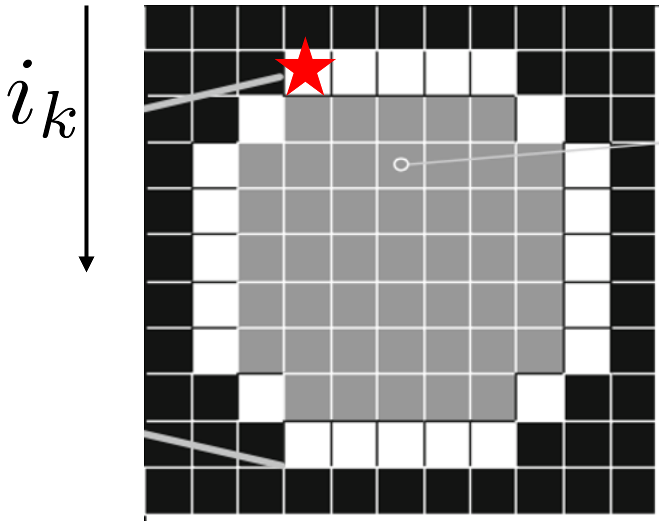
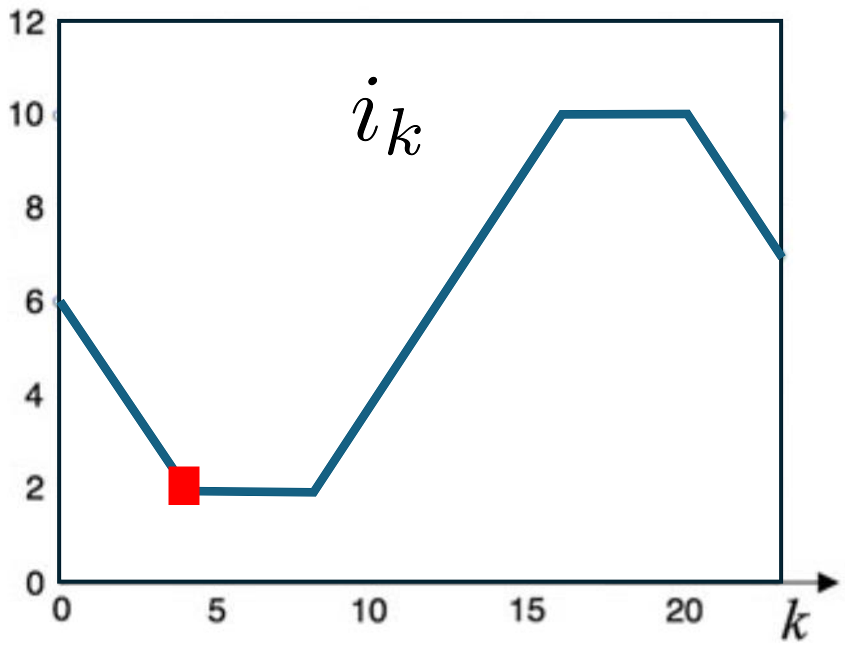
k
i_k


i_k


i_k
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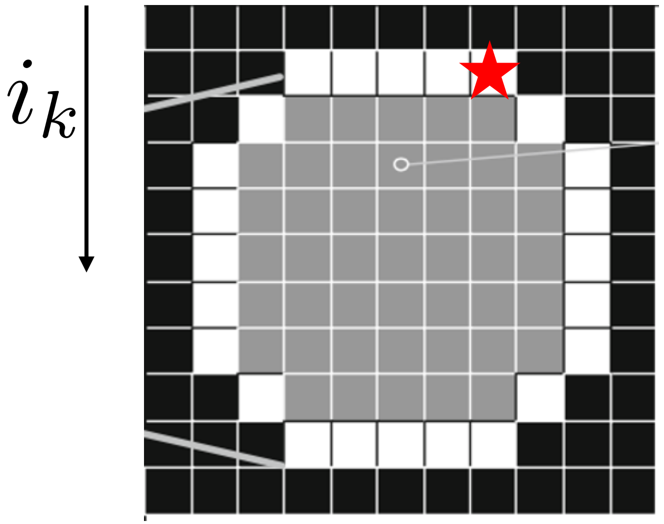
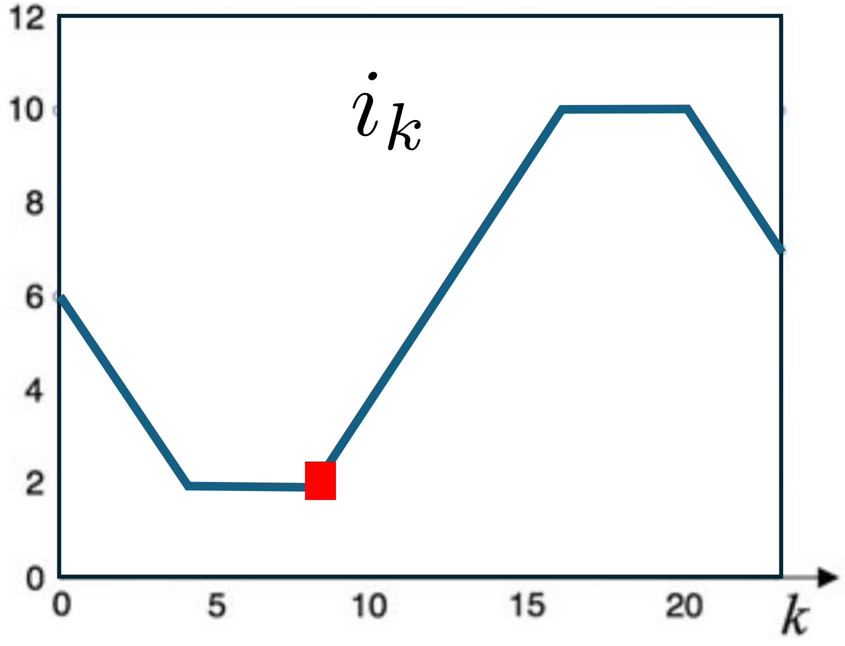
k


i_k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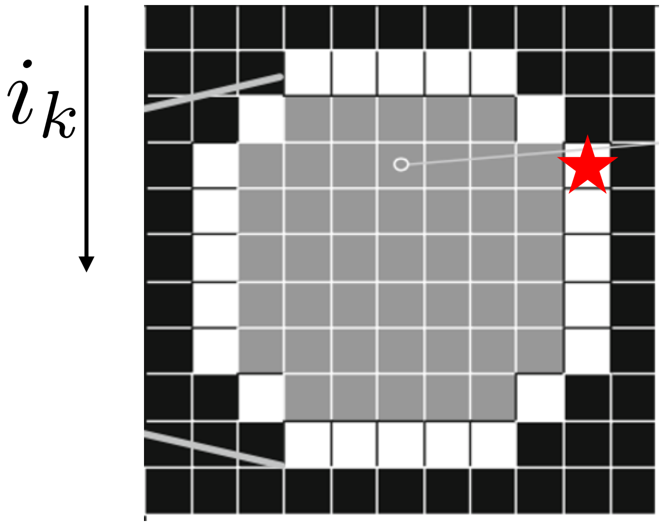
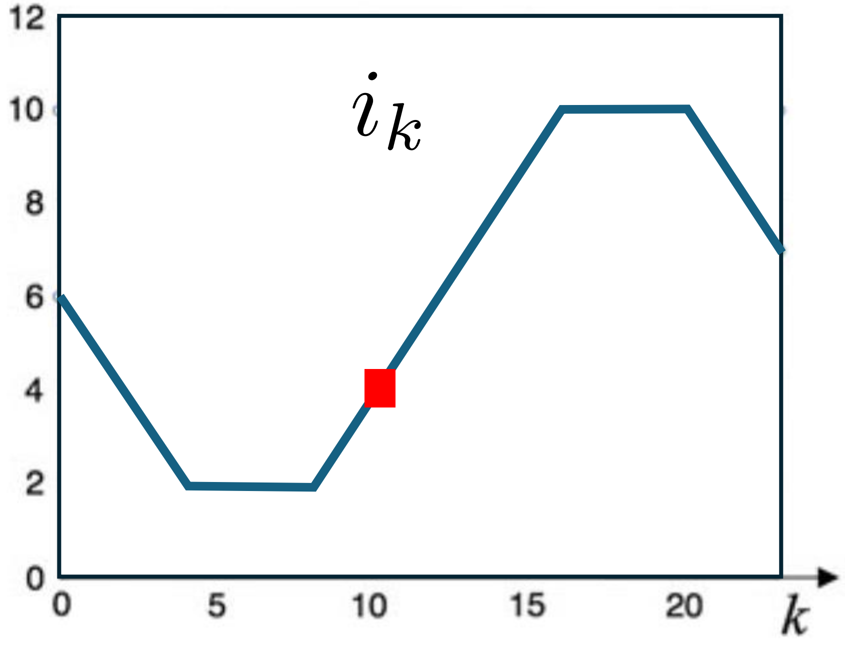
k


i_k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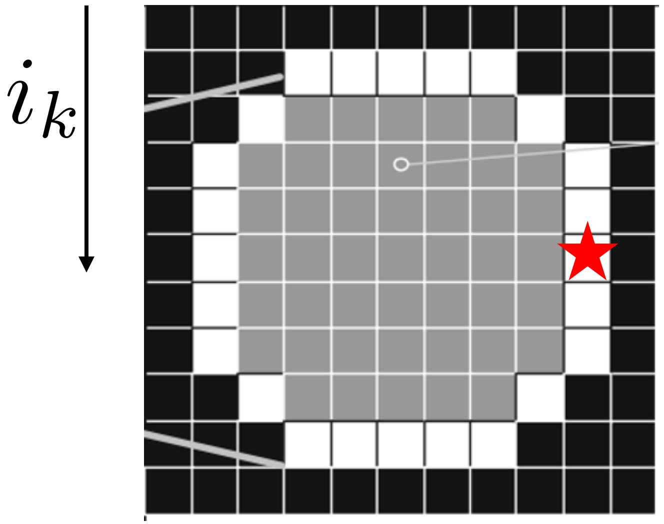
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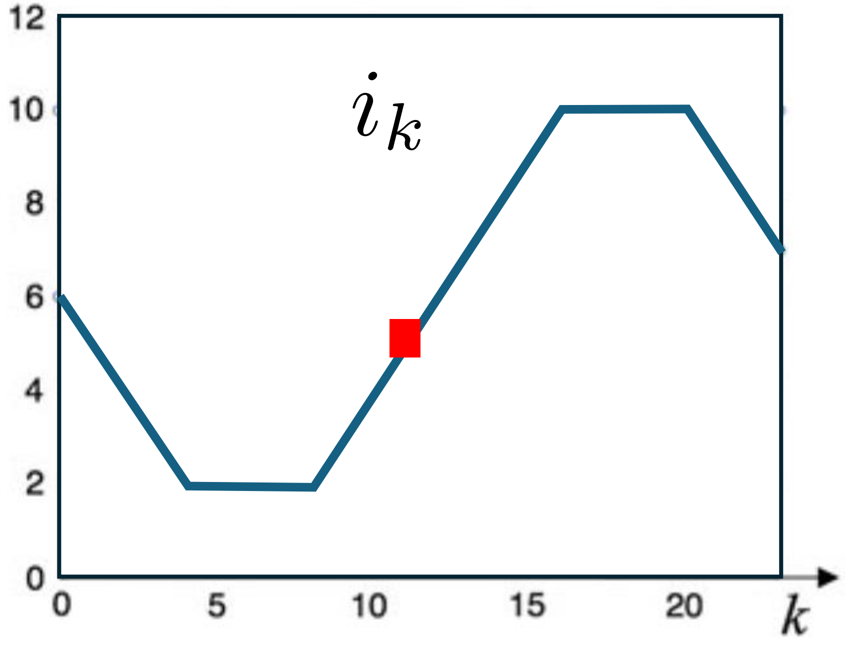
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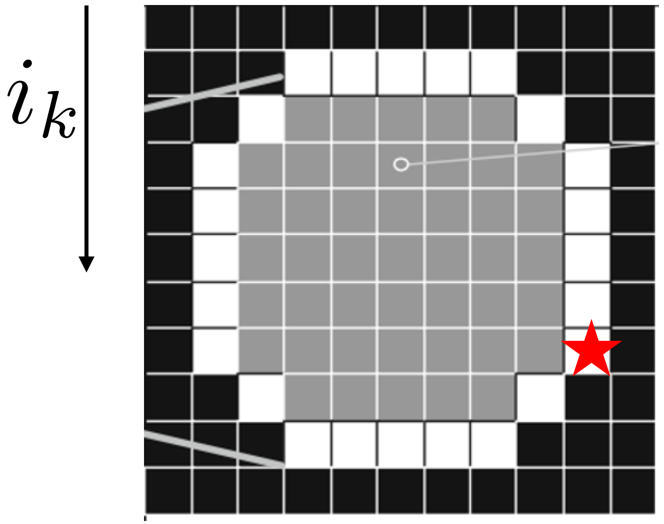
i_k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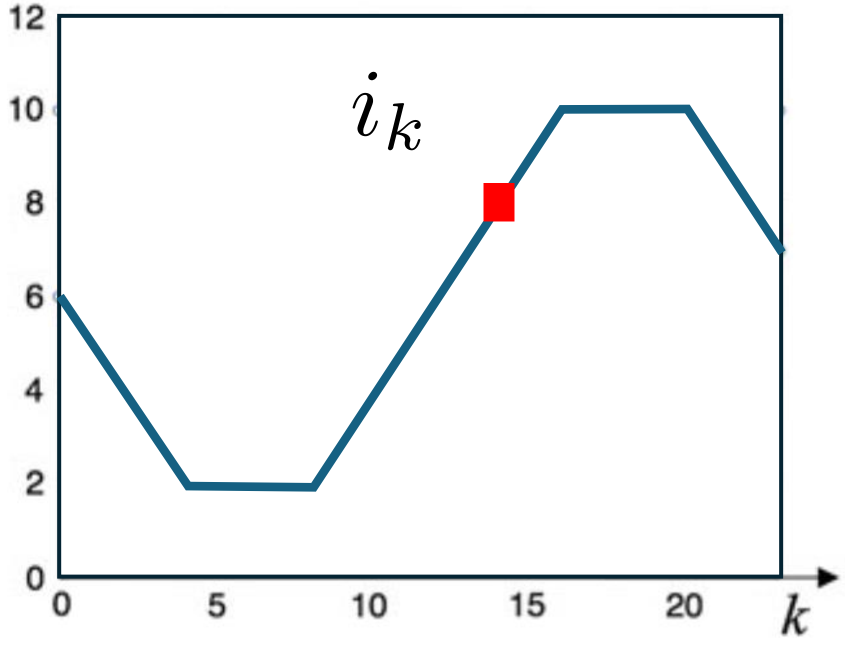
i_k
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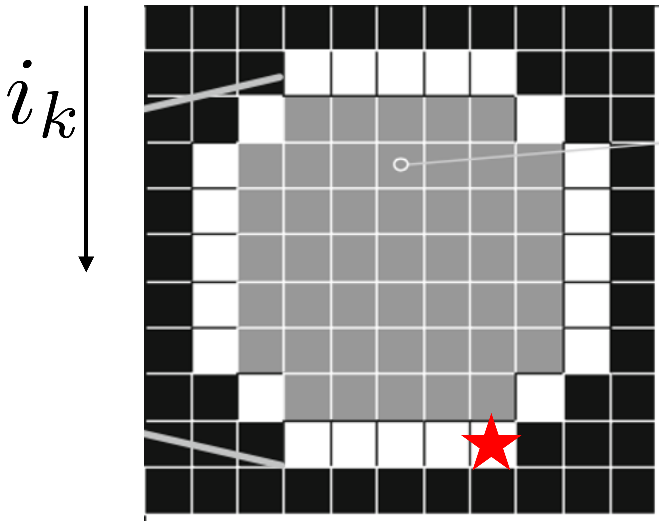
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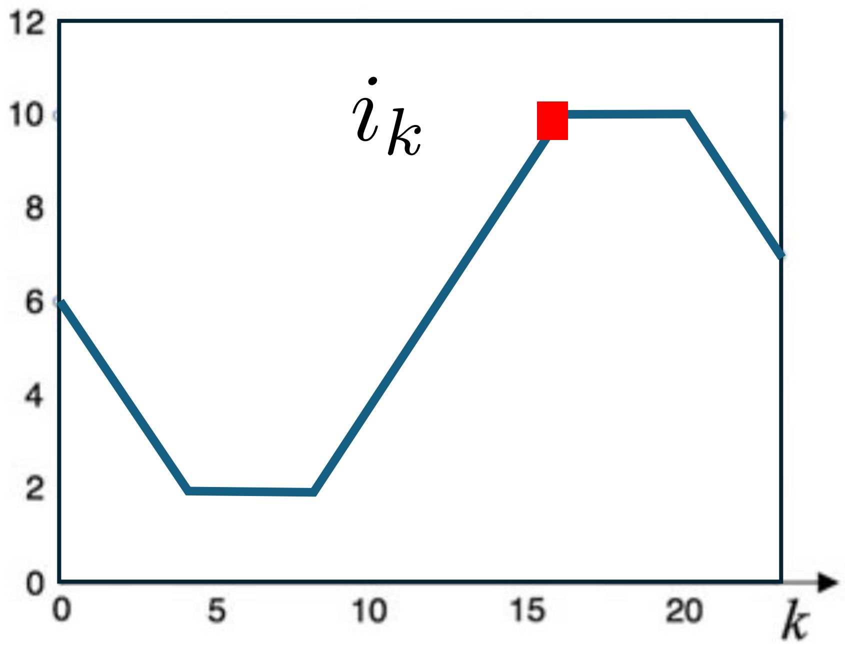
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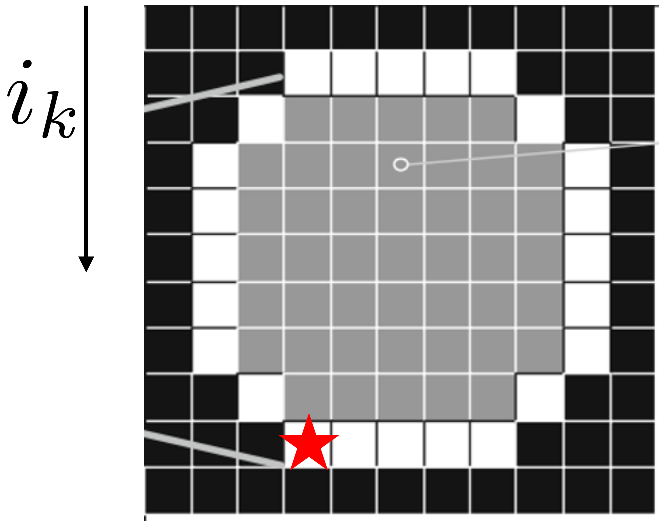
i_k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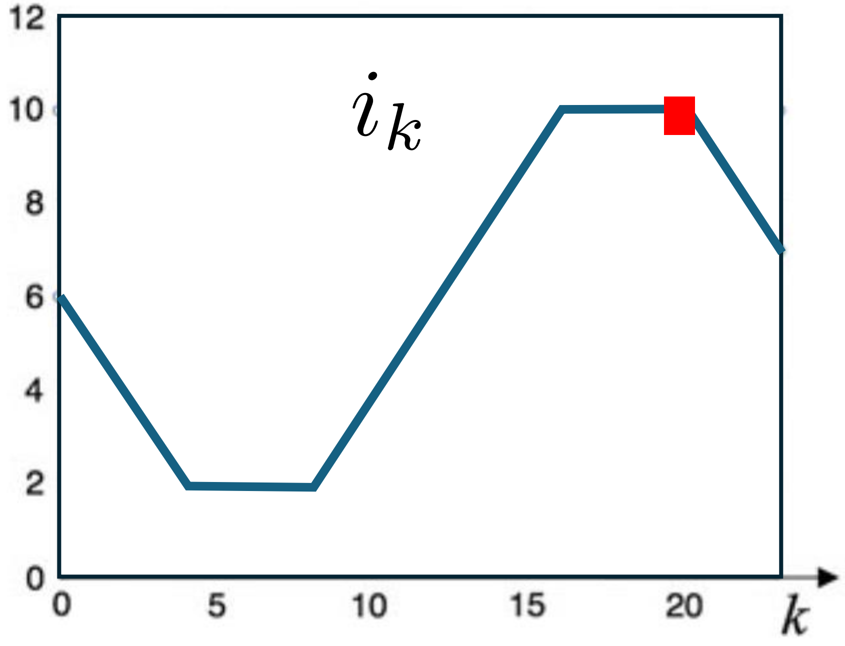
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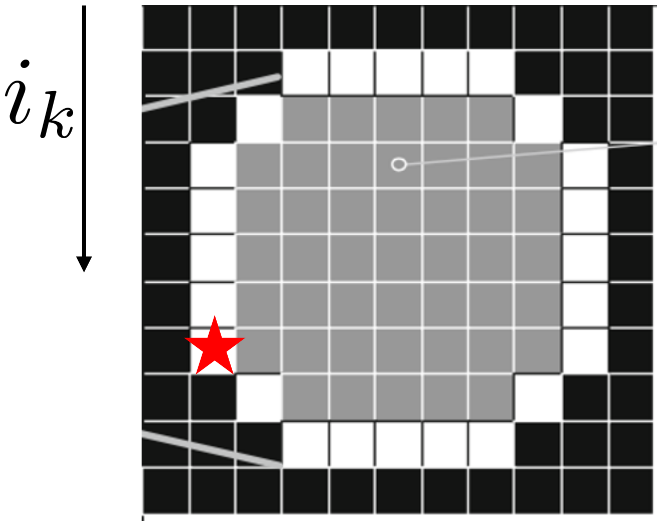
i_k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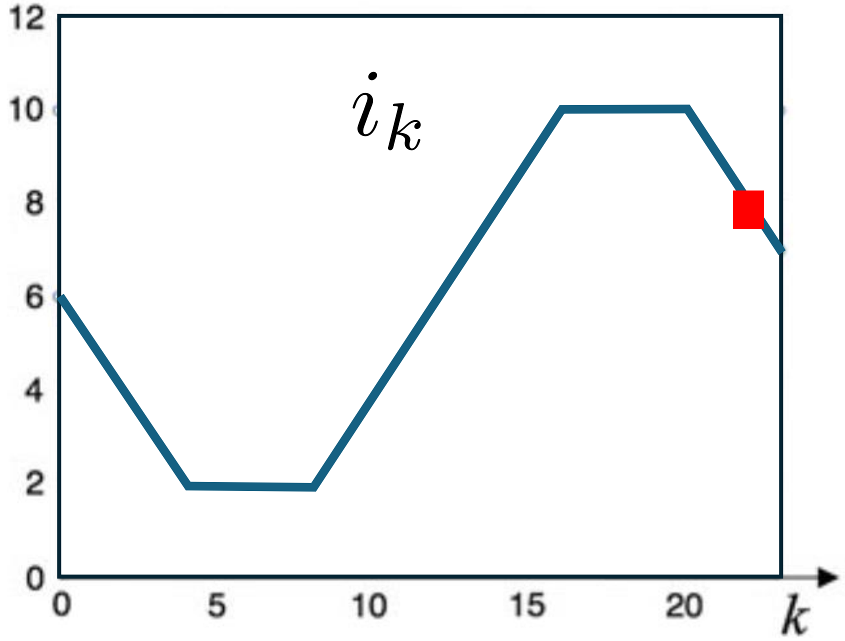
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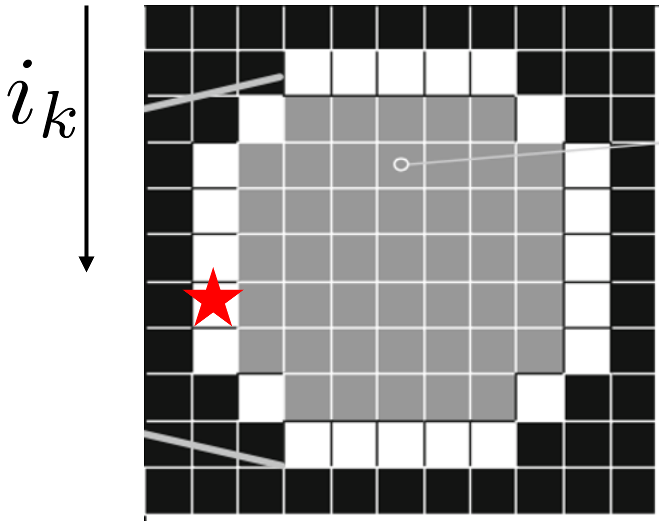
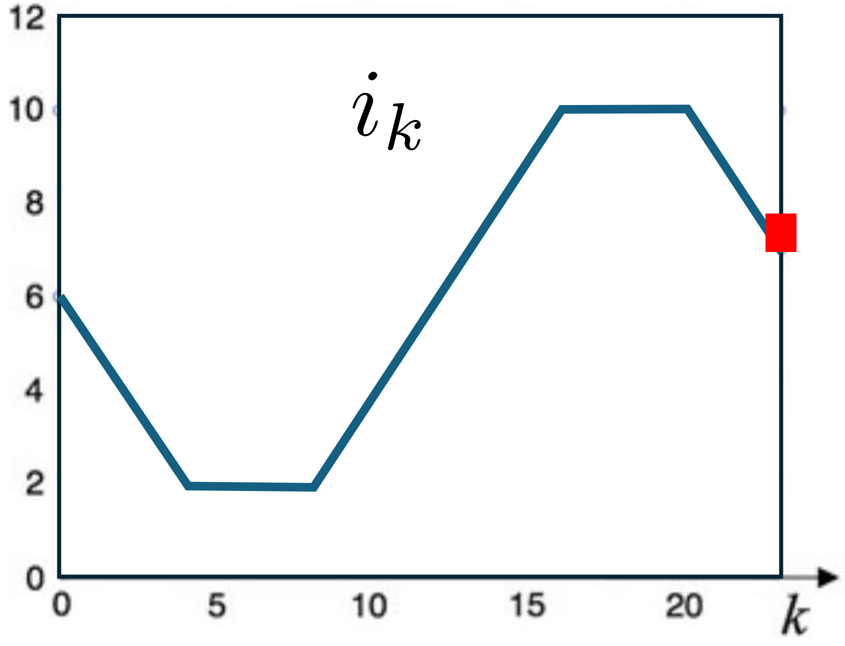
k
i_k



i_k



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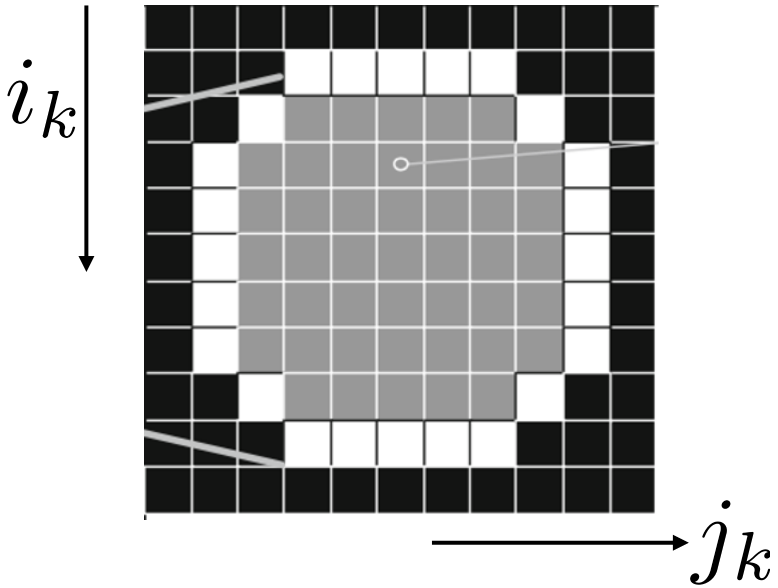
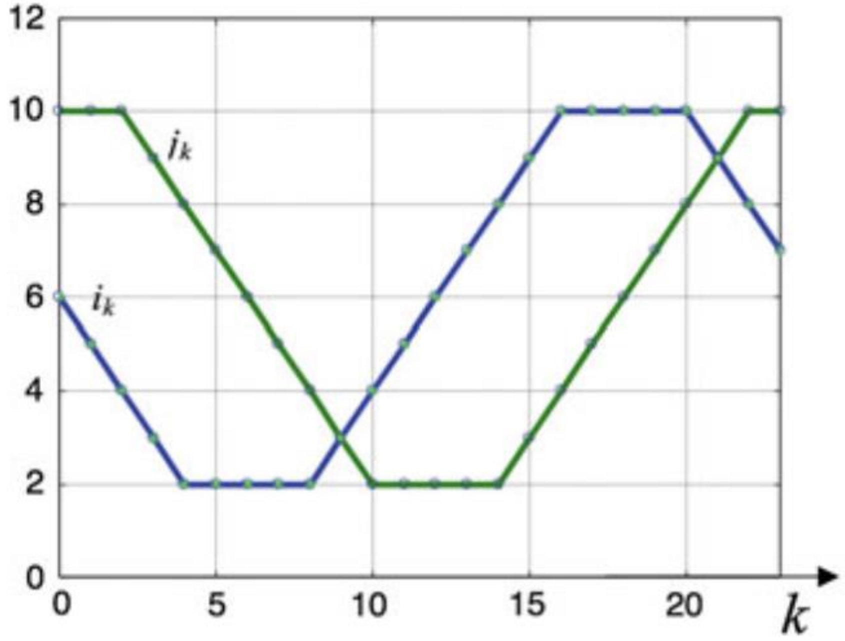
i_k


i_k
j_k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2
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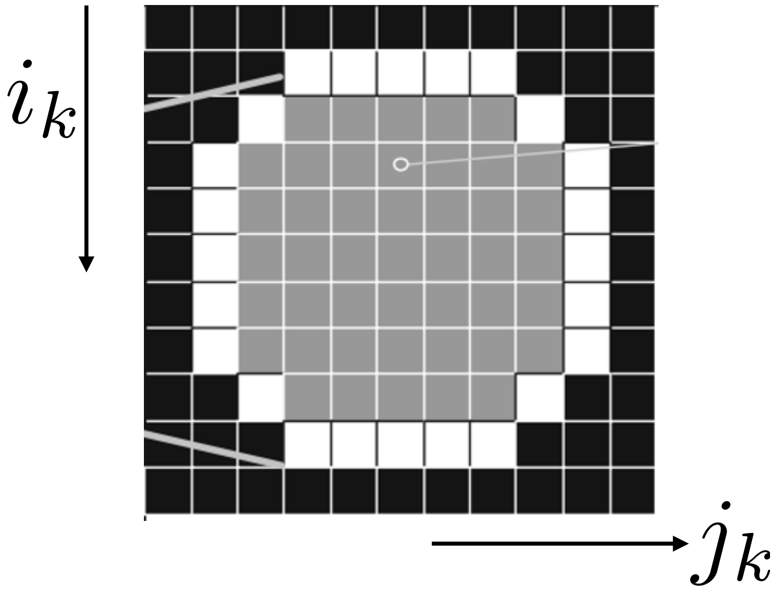
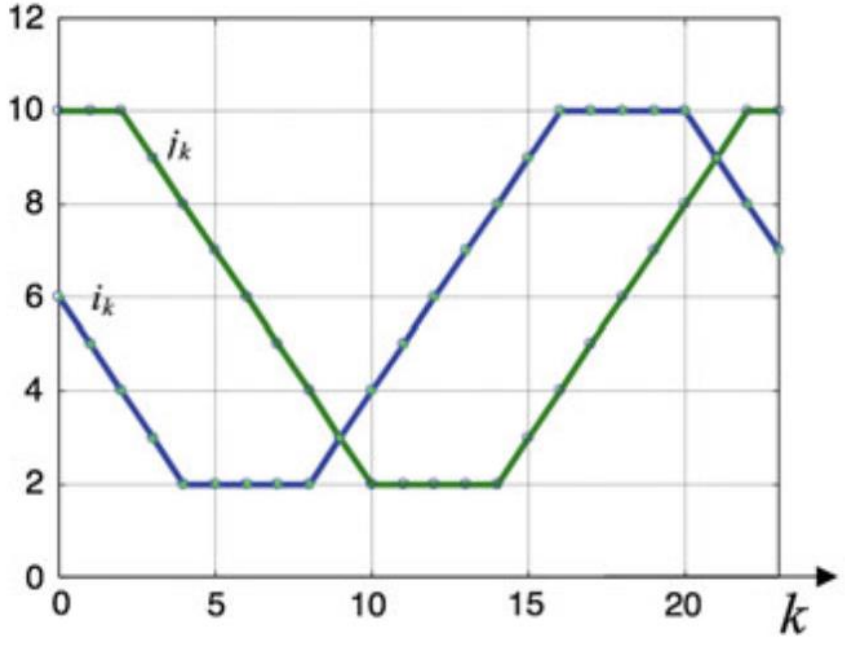
5
10
15
20
k
j_k
i_k



for $n = 0, \ldots, L-1$.

$$F_n = \sum_{k=0}^{L-1} (i_k + j \cdot j_k) e^{-j\frac{2\pi kn}{L}}$$



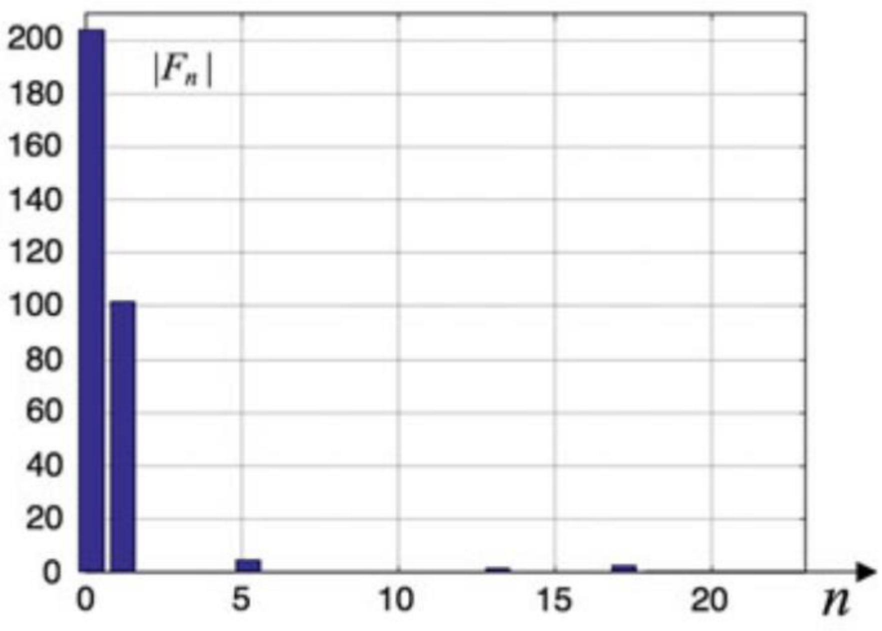


Fig. 5.4 Coordinates of the boundary of region of Fig. 5.1 and the Fourier descriptors

Ejemplo

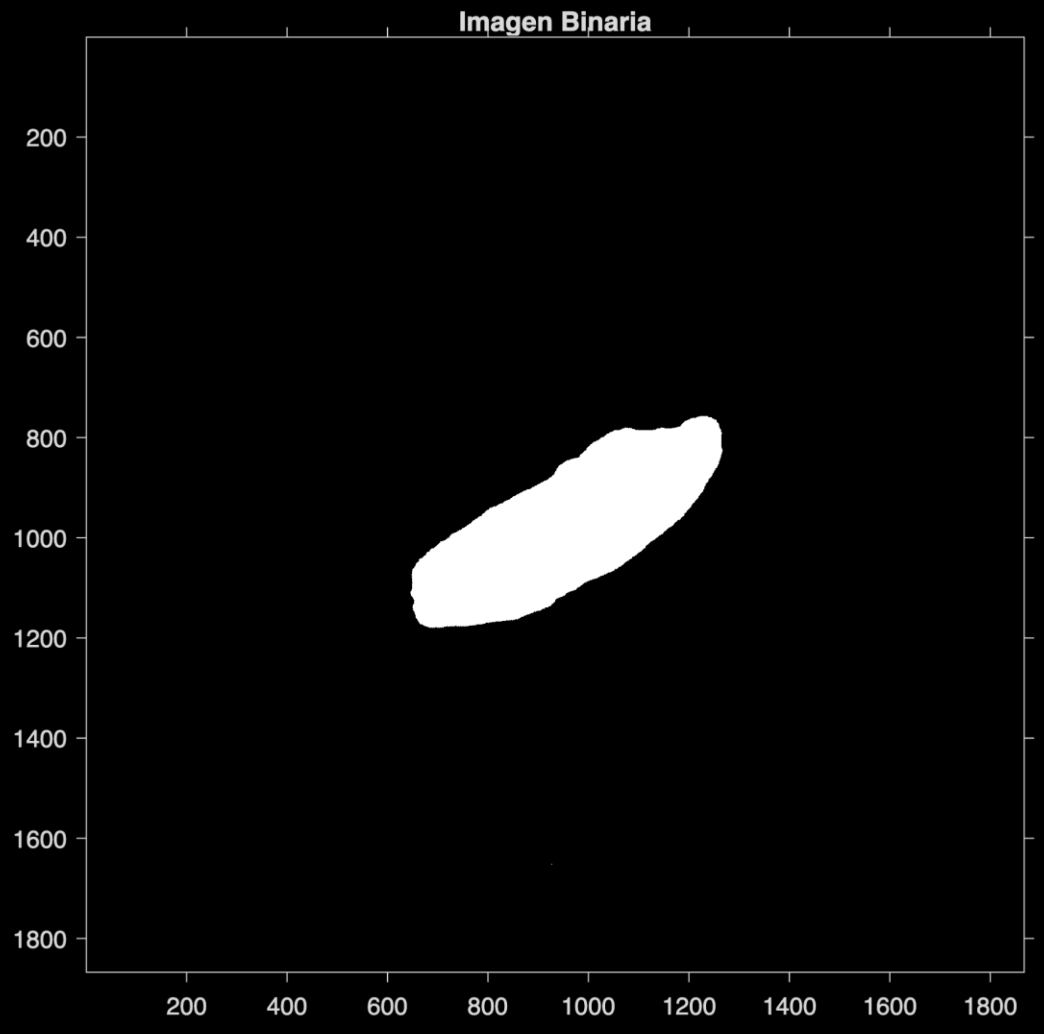


Imagen Binaria
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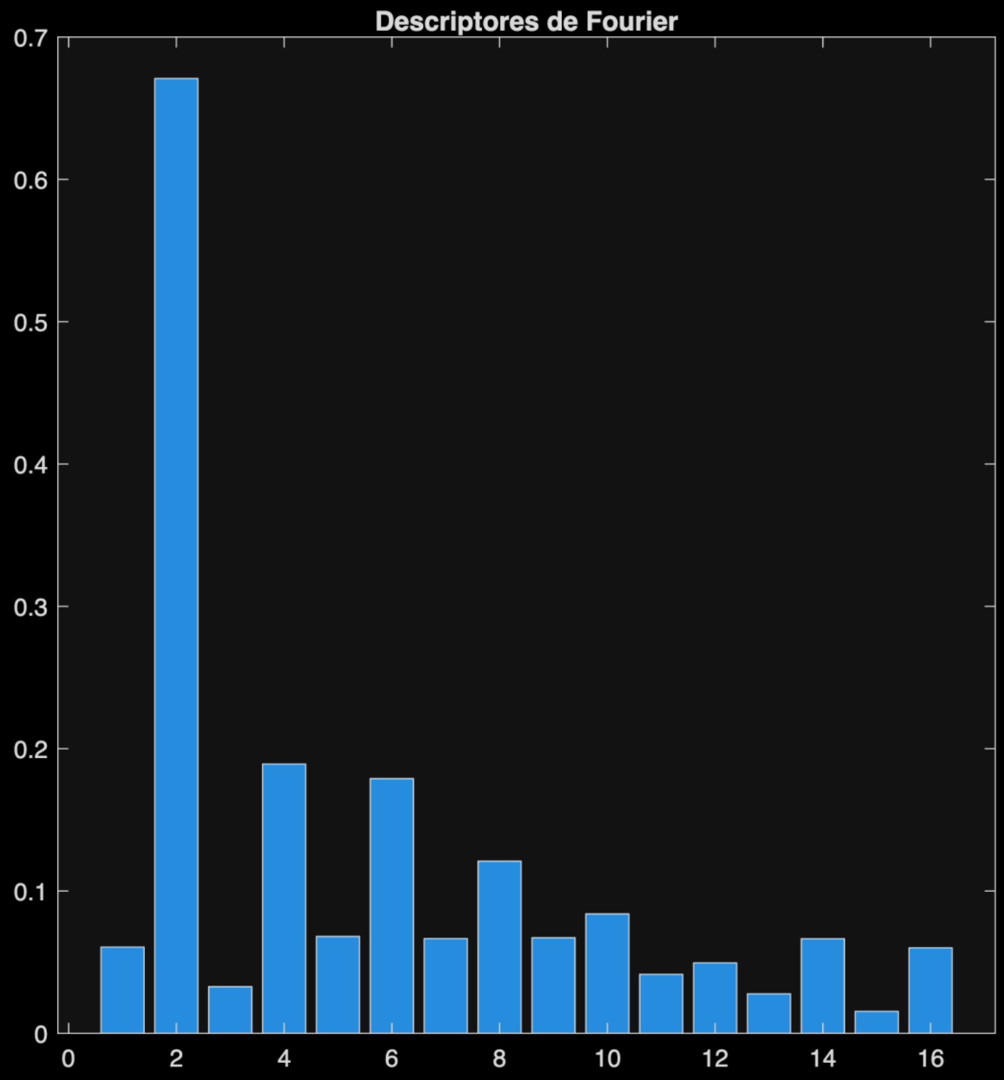

Descriptores de Fourier
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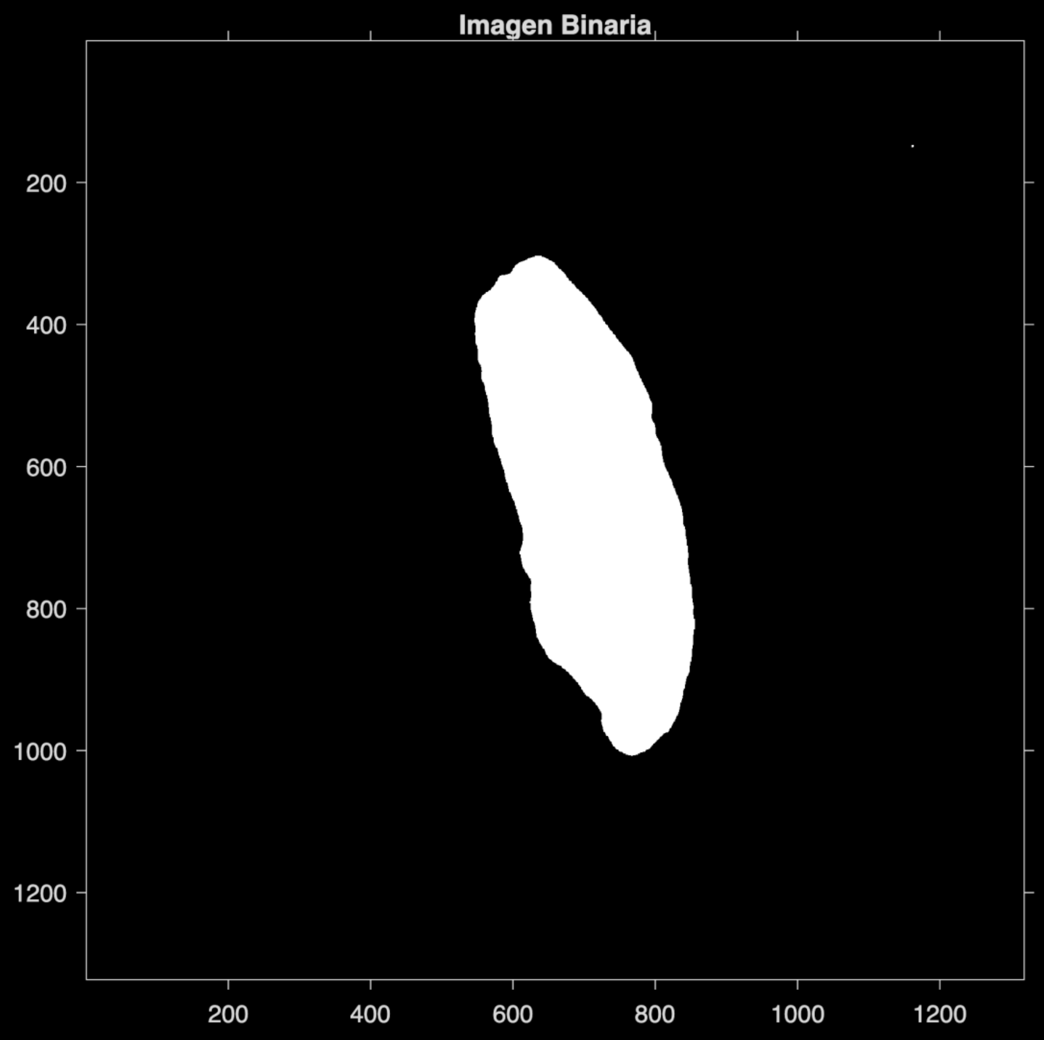
0.6
0.5
0.4
0.3
0.2
0.1
0
2
4
6
8
10
12
14
16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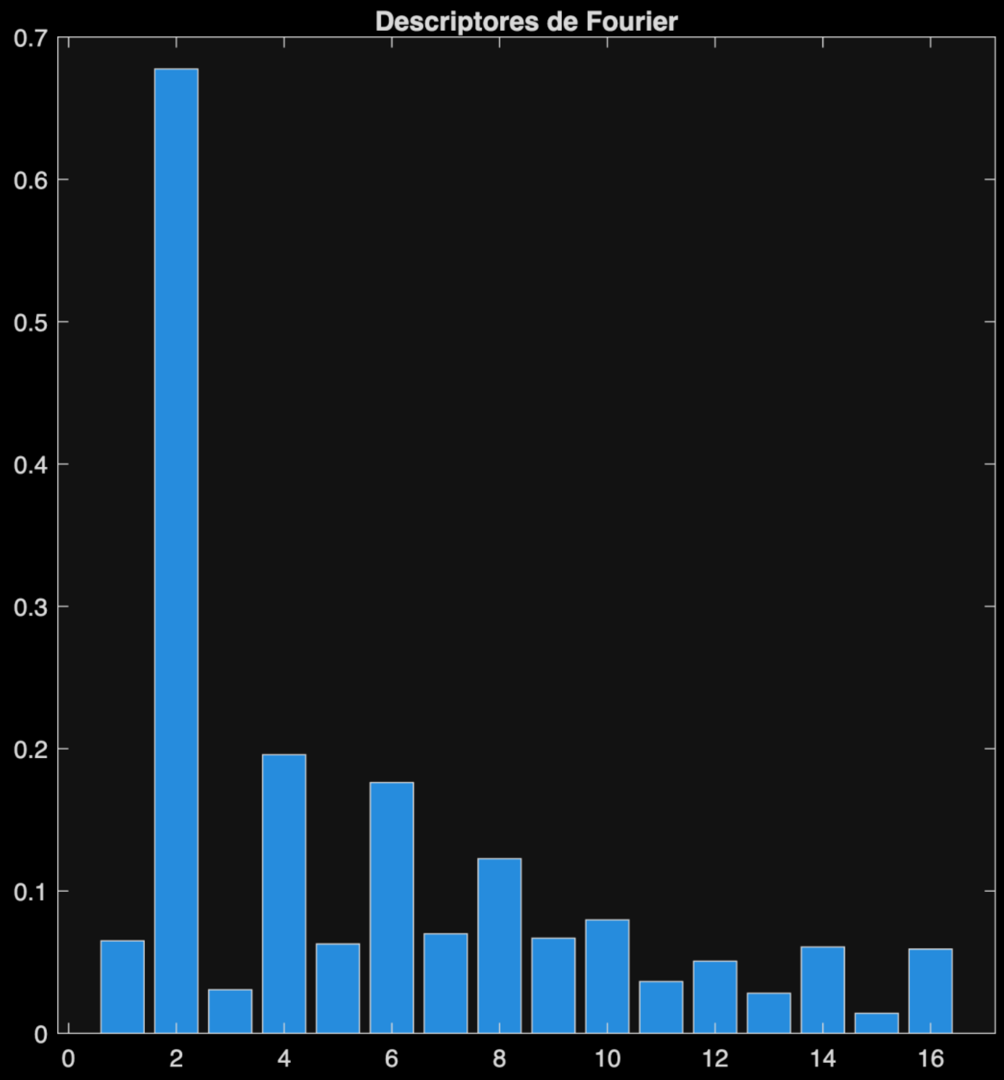
Descriptores de Fourier
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0.3
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2
4
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10
12
14
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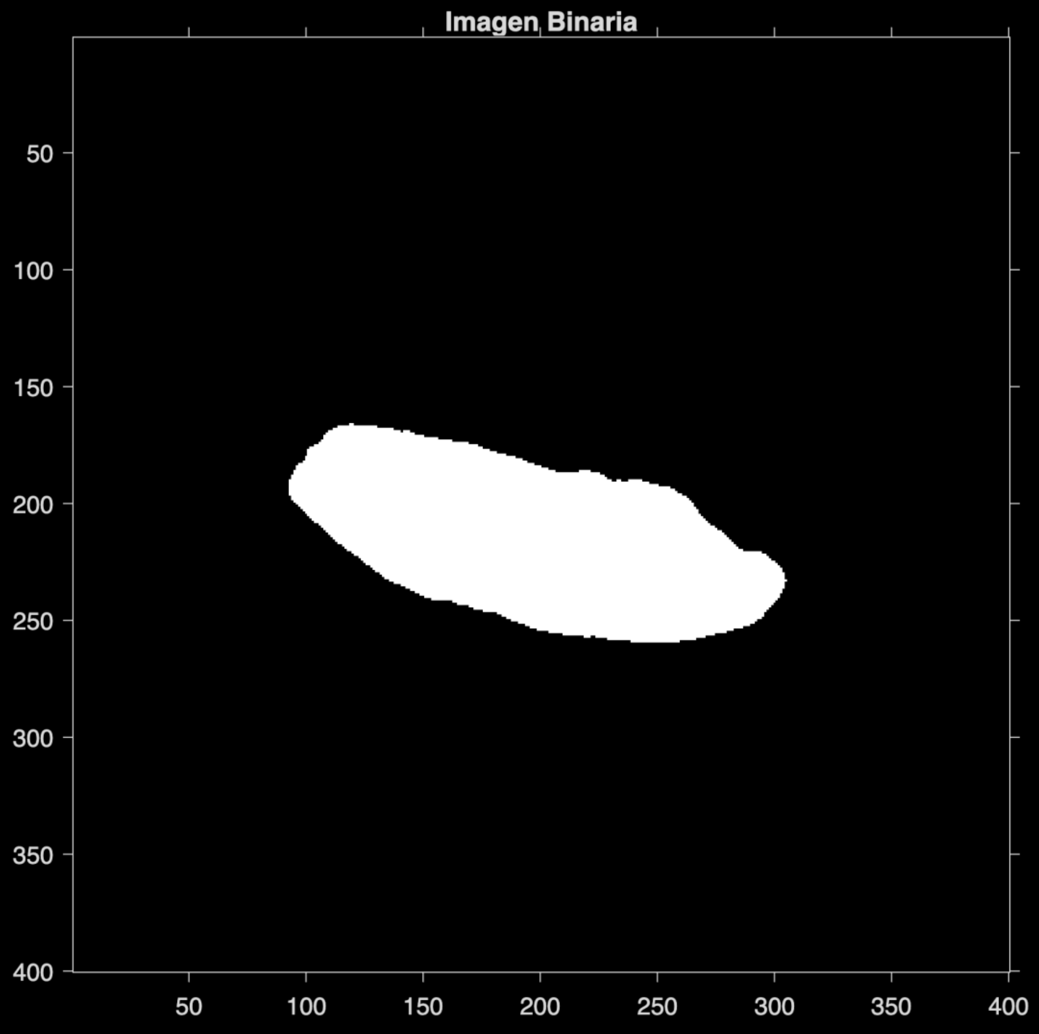
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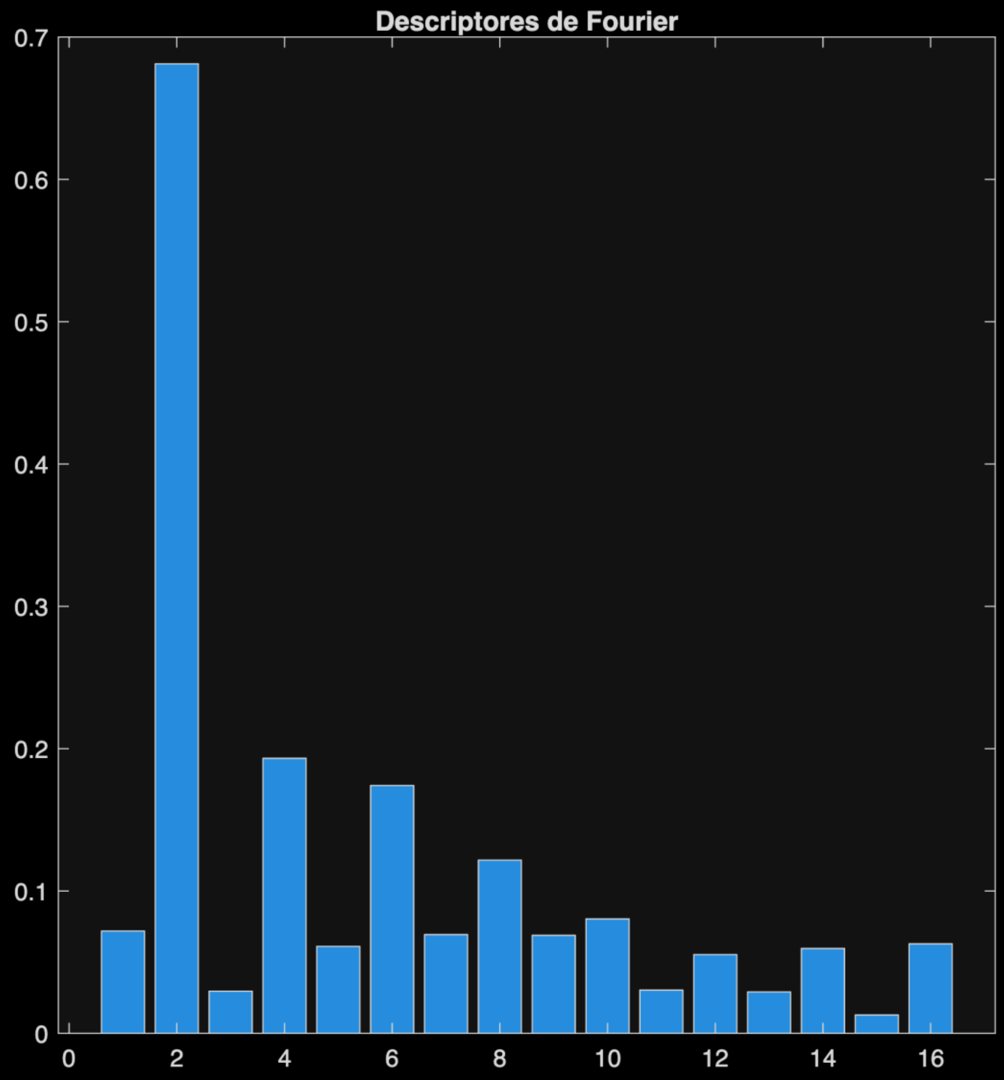
Descriptores de Fourier
0.7
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0.3
0.2
0.1
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2
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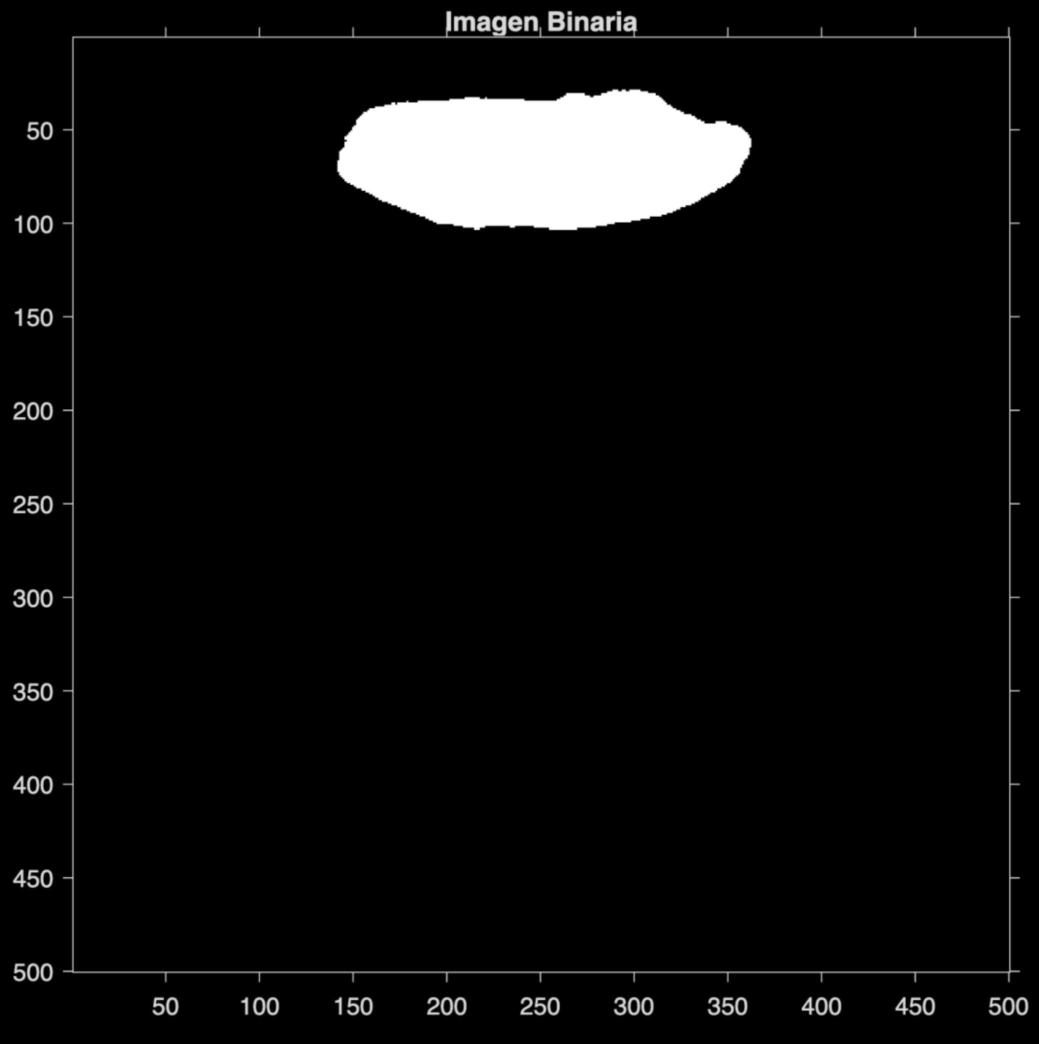
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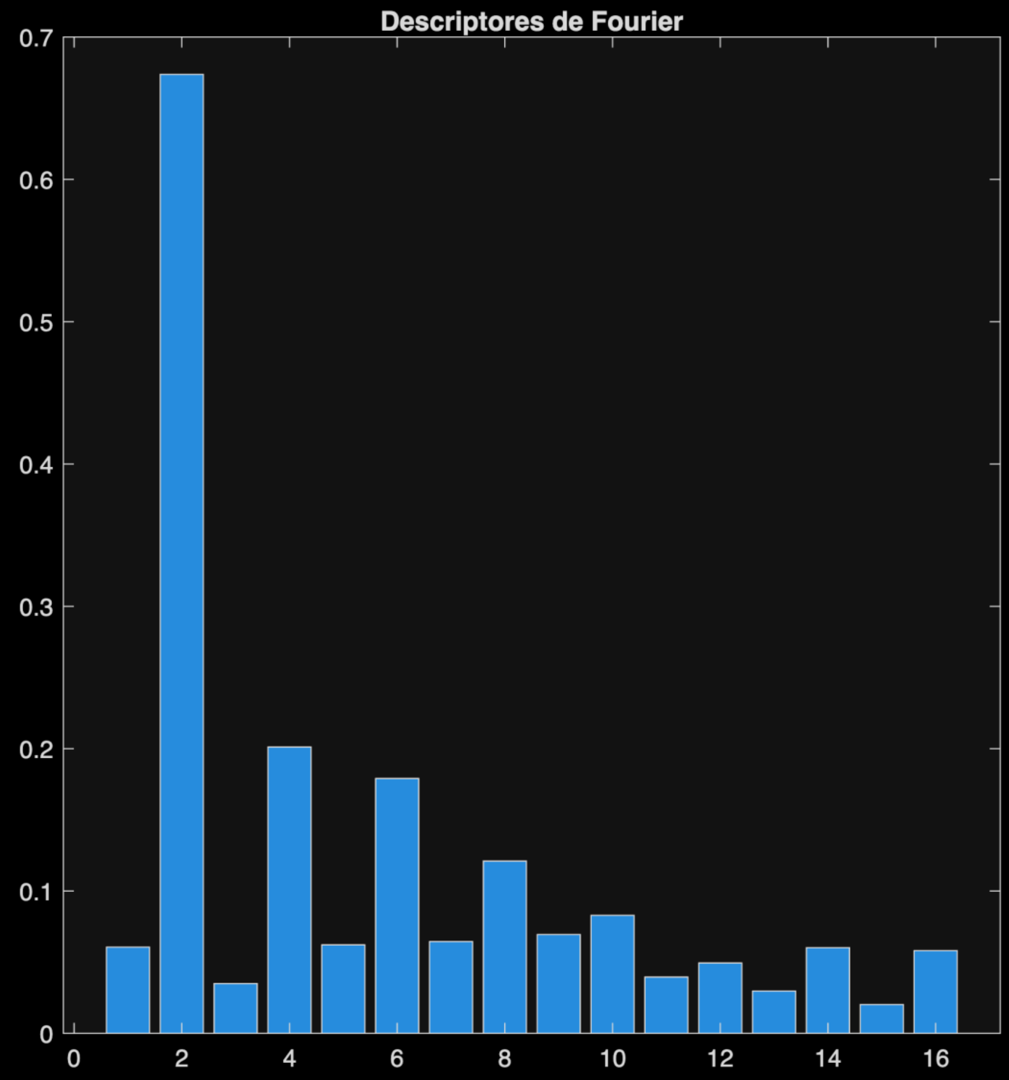
Descriptores de Fourier
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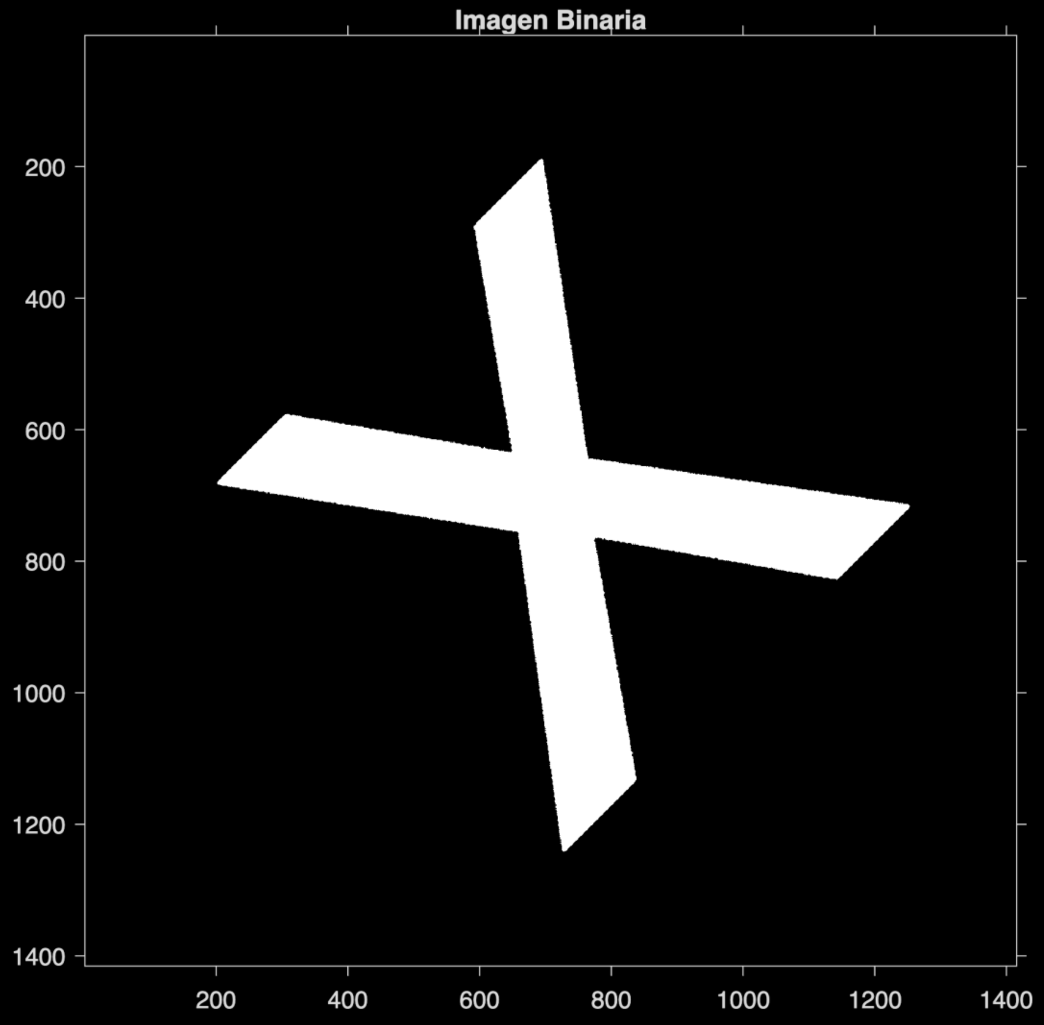
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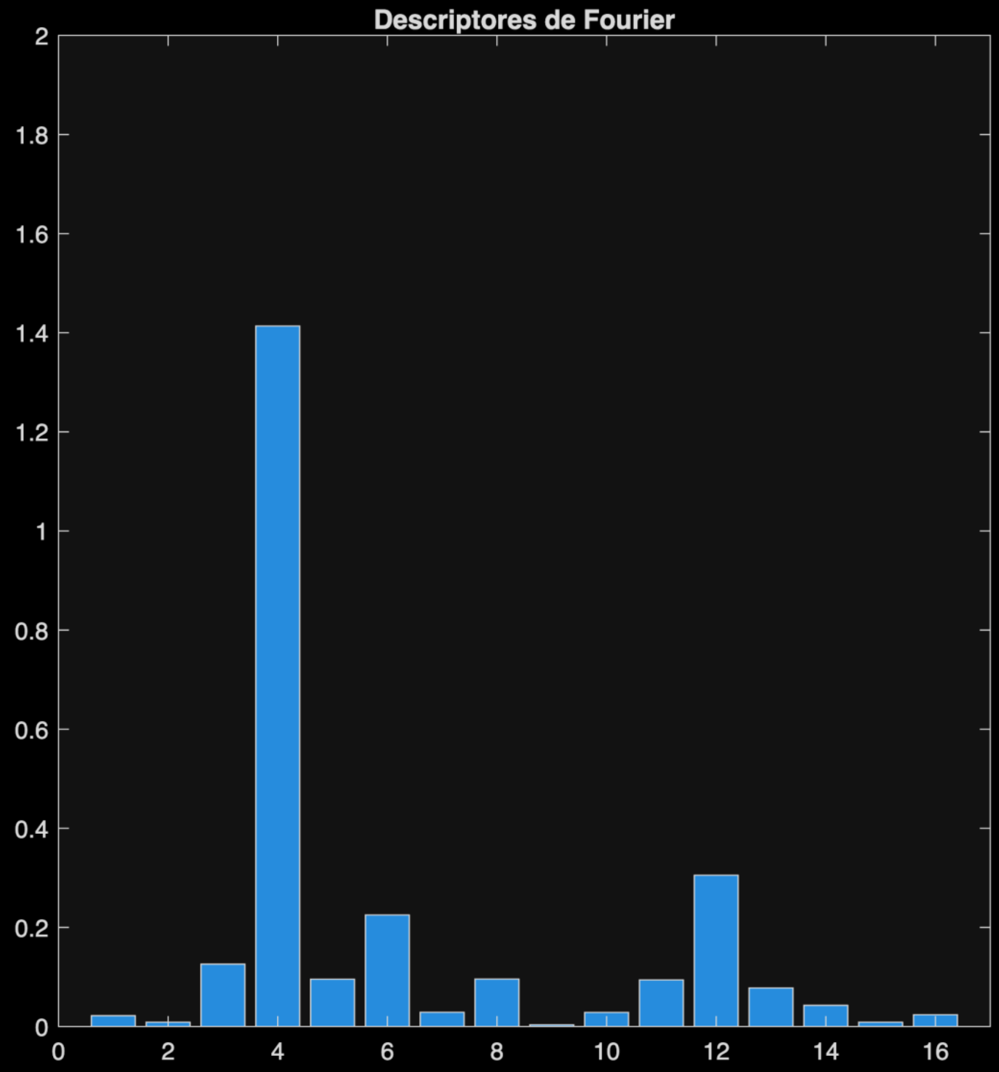
Descriptores de Fourier
0
0.2
0.4
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1
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1.4
1.6
1.8
2
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8
10
12
14
16

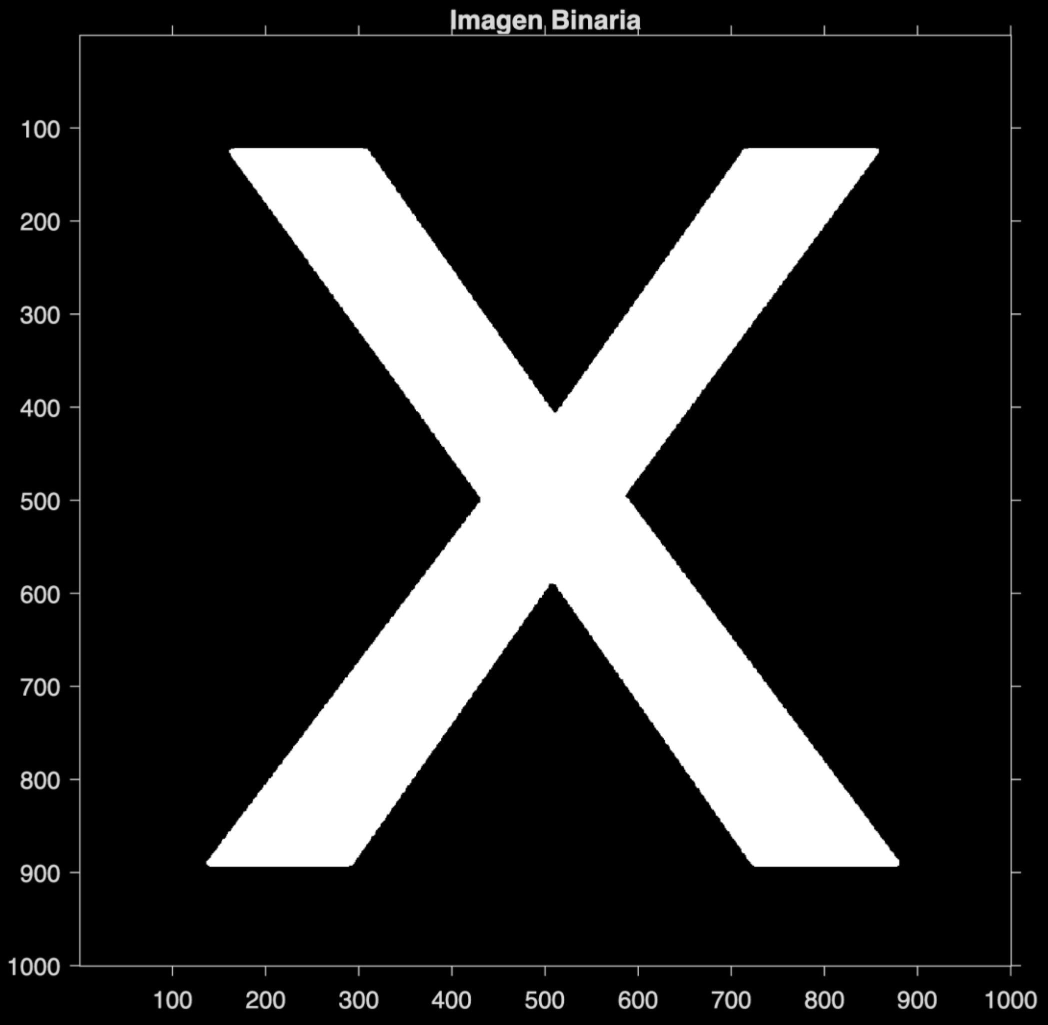
Imagen Binaria
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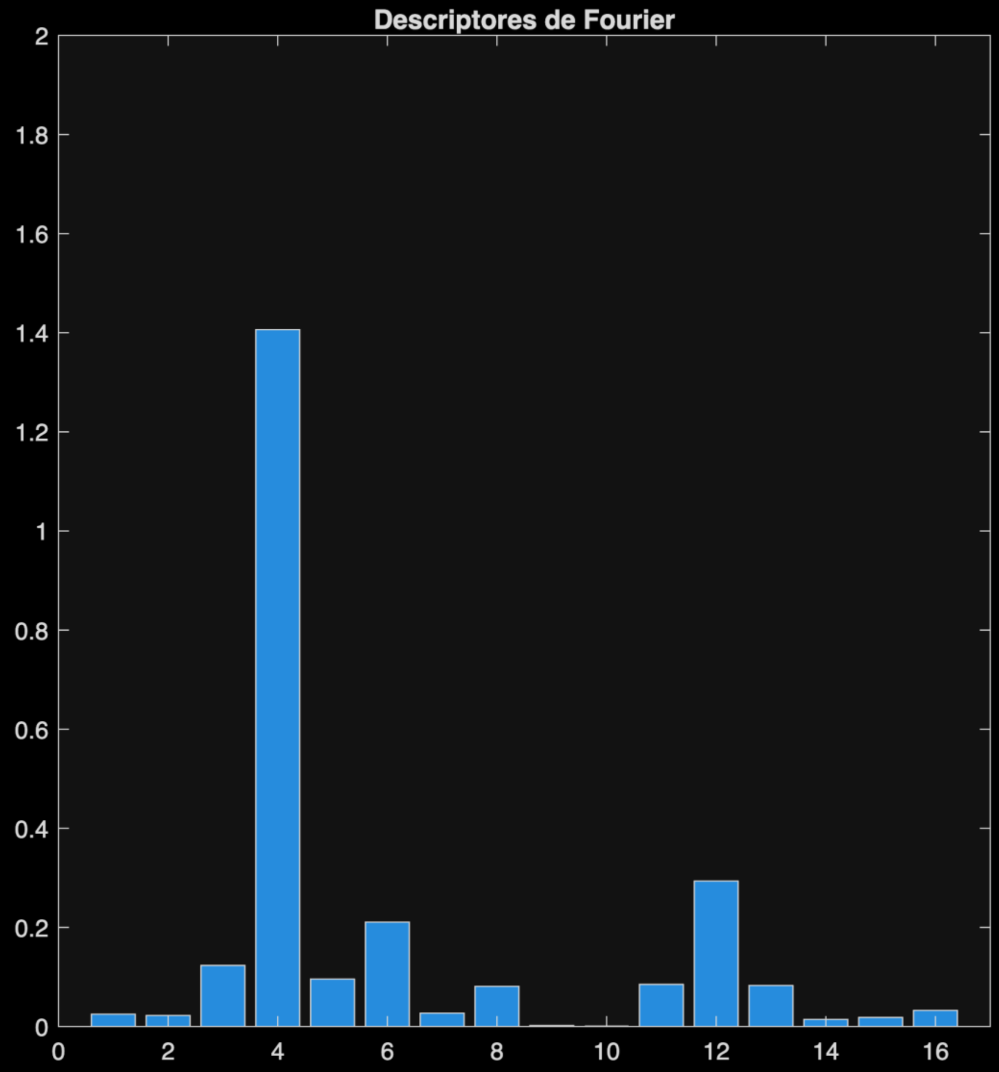
Descriptores de Fourier
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0.8
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1.4
1.6
1.8
2
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2
4
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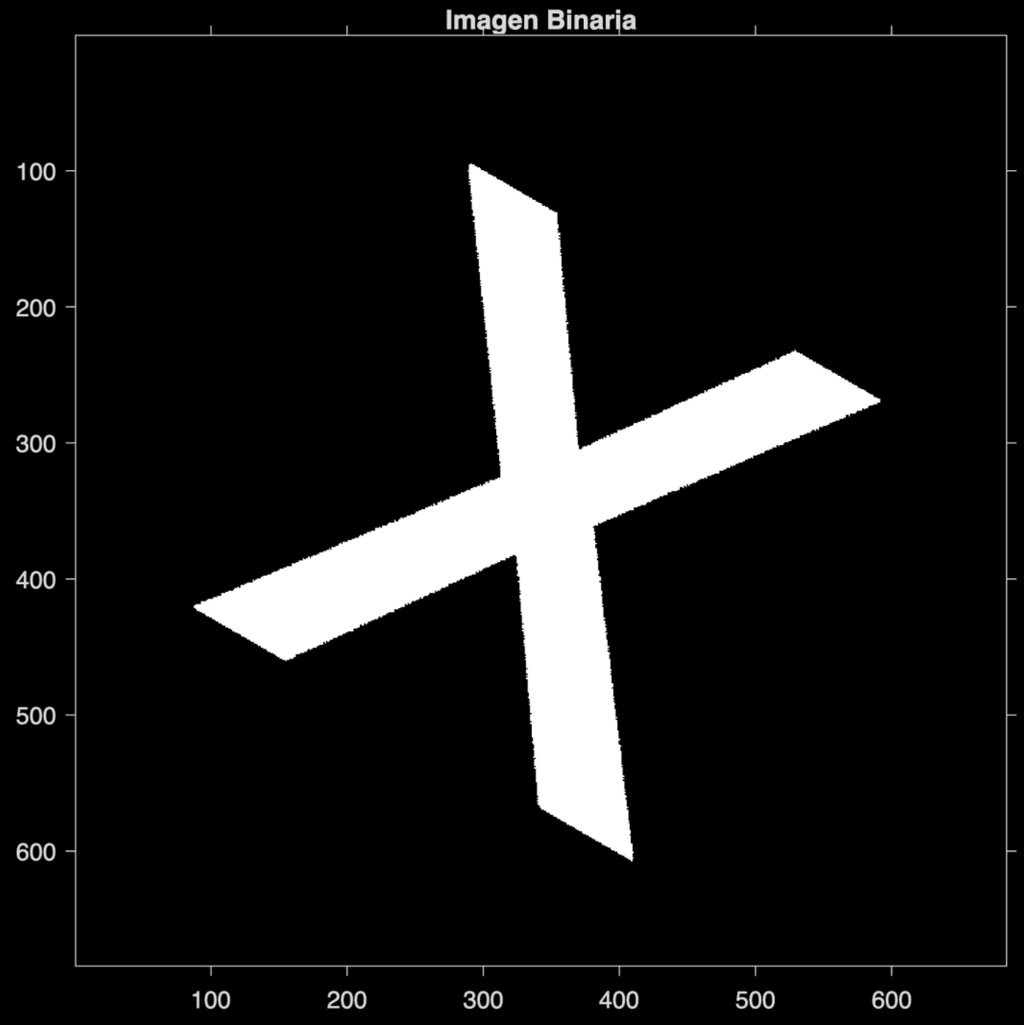
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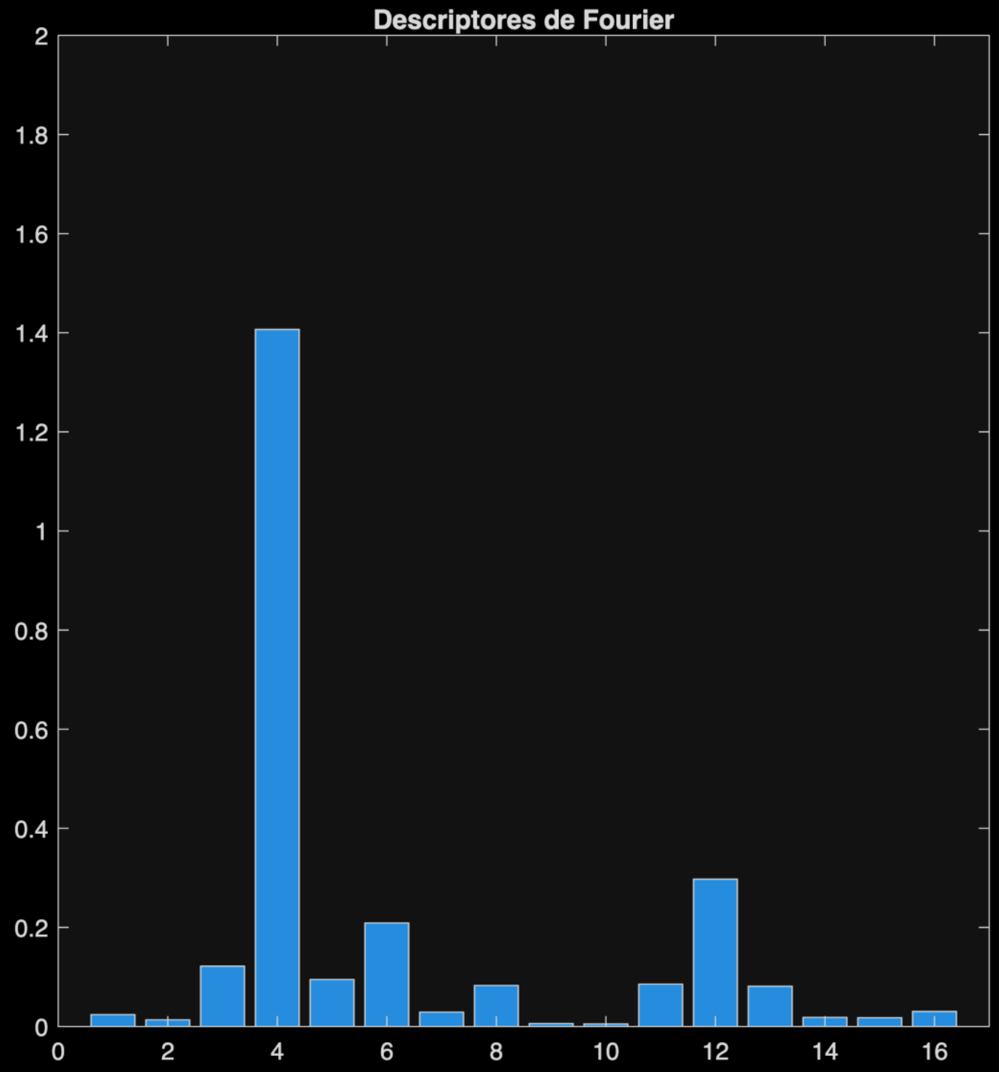
Descriptores de Fourier
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1.6
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1.2
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0.6
0.4
0.2
0
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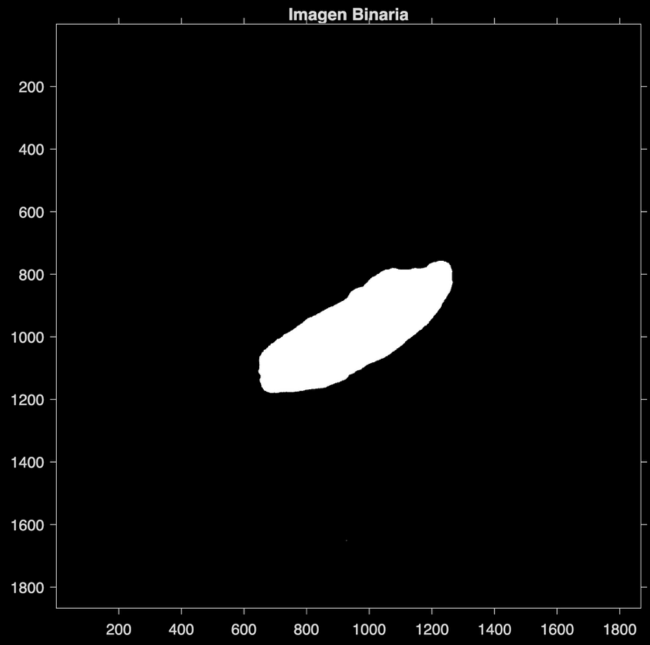
8
10
12
14
16



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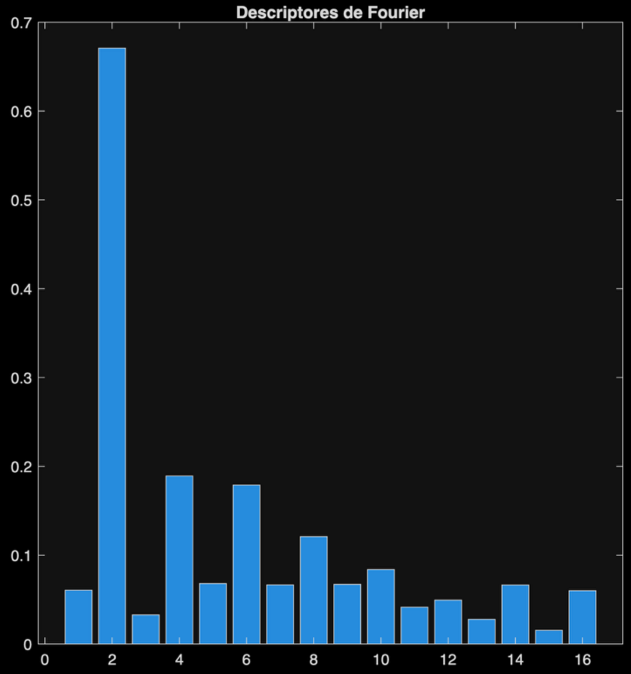

Descriptores de Fourier
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0.4
0.5
0.6
0.7
2
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10
12
14
16

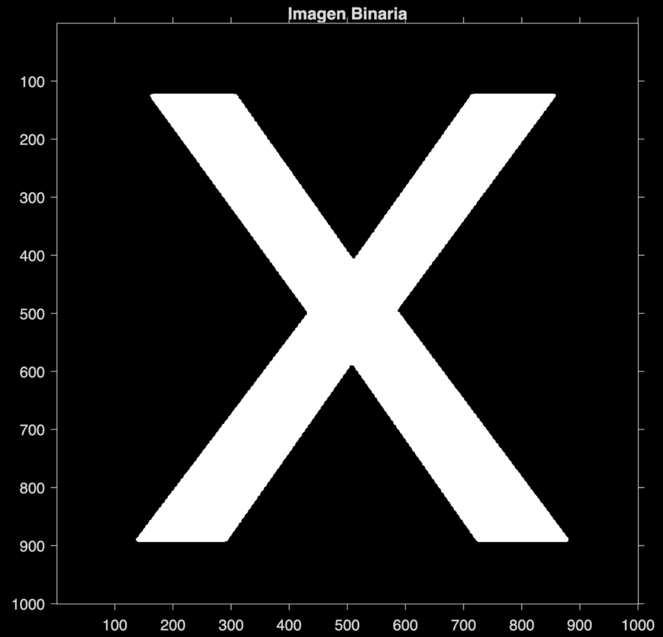

Imagen Binaria
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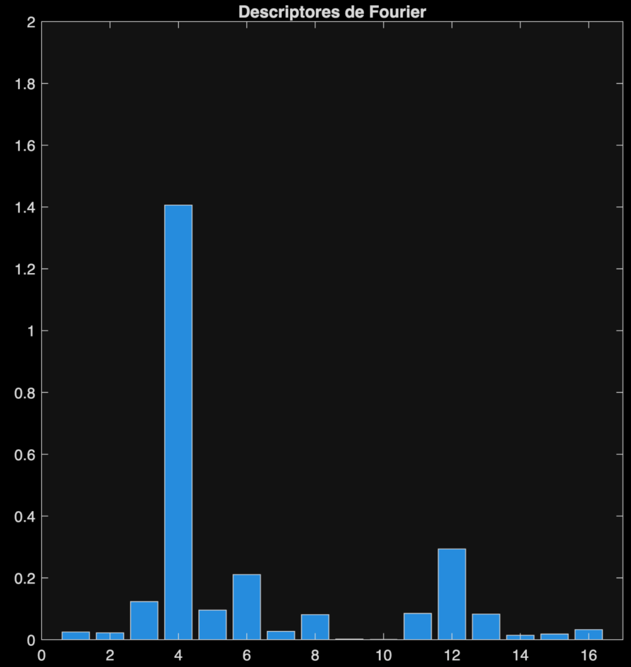

Descriptores de Fourier
0
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0.6
0.8
1
1.2
1.4
1.6
1.8
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12
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